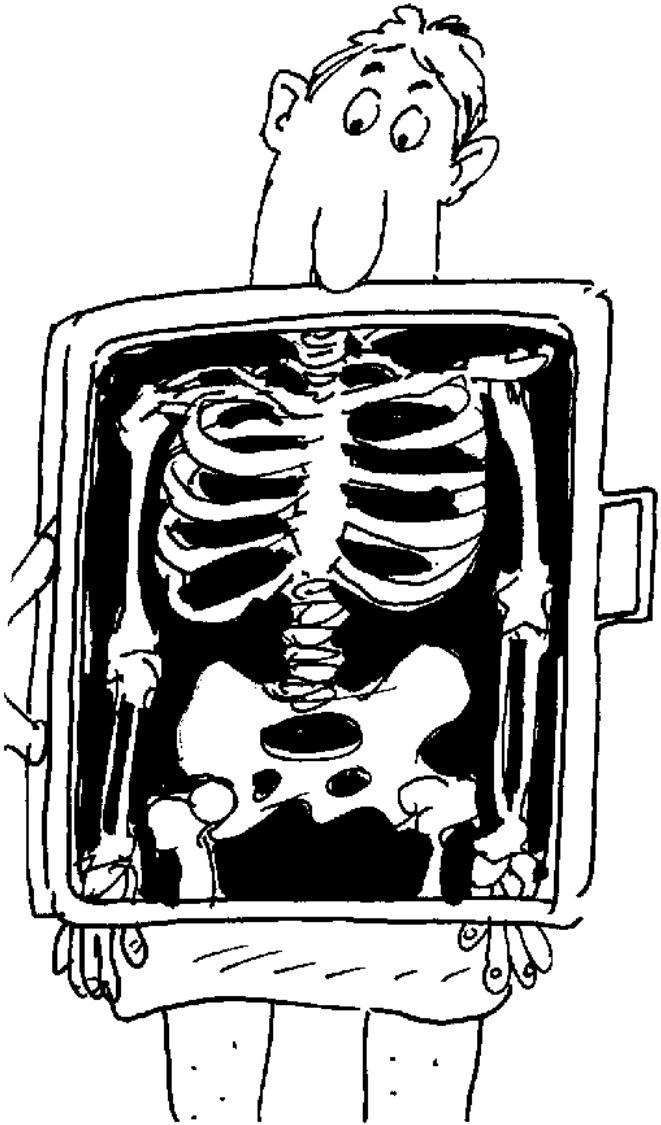
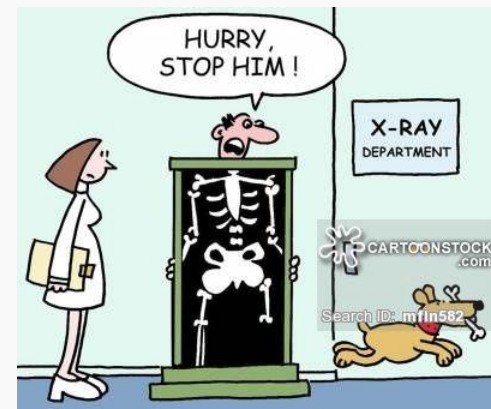


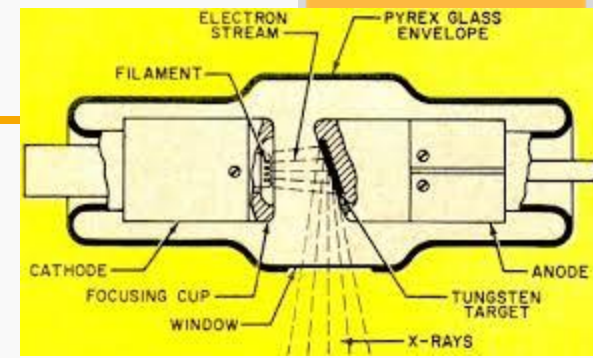


Chest Radiography: The Basics

Joseph Mazzie D.O.
Department of Clinical Specialties,
Radiology

- Understand basic principles of x-rays
- Understand basic normal chest radiographic anatomy
- Formulate an approach to evaluating and reading chest radiographs





- An x-ray beam is generated by passing an electron beam through a vacuum between a cathode (-) and an anode (+).

- The positively charged anode attracts the rapidly moving, negatively charged electrons.
- As the electrons collide and interact with the atoms on the anode target, a great amount of energy is produced. 1% of this energy is in the form of x-radiation. 99% appears as heat and must be removed from the anode.
- The tube enclosure is shielded and a series of lead shutters allow the diagnostic beam to exit.



- Scientific unit of measurement for whole body radiation dose, called "effective dose," is the **millisievert (mSv)**.
- We use "**effective dose**" when they talk about the risk of radiation to the entire body. Risk refers to Radiation Dose in X-Ray and CT Exams refers to possible side effects, such as the chance of developing a cancer later in life.
- Effective dose takes into account how sensitive different tissues are to radiation. If you have an x-ray exam that includes tissues or organs that are more sensitive to radiation, your effective dose will be higher.

- According to recent estimates, the average person in the U.S. receives an effective dose of about 3 mSv per year from natural radiation, which includes cosmic radiation from outer space.
- These natural "background doses" vary according to where you live.
 - i.e. high altitudes such as Colorado or New Mexico receive about 1.5 mSv more per year than those living near sea level.
- A coast-to-coast round trip airline flight is about 0.03 mSv due to exposure to cosmic rays.
- The largest source of background radiation comes from radon gas in our homes (about 2 mSv per year) and that amount varies widely depending on where you live.
- Simply put, the amount of radiation from **one adult chest x-ray (0.1 mSv) is about the same as 10 days of natural background radiation that we are all exposed to as part of our daily living**
- <https://www.xrayrisk.com/calculator/calculator.php>

- Basics

- As the x-rays pass through an object, based on the density of that object the x-ray(s) will either be “absorbed” (and/or scattered)..
- It is the degree of “absorption” of the x-ray which creates the image we see on the film

Darkest

Minimum x-ray

**Transmission
(least dense
tissue)**

air

fat

soft tissue/fluid*

calcium

bone

x-ray contrast

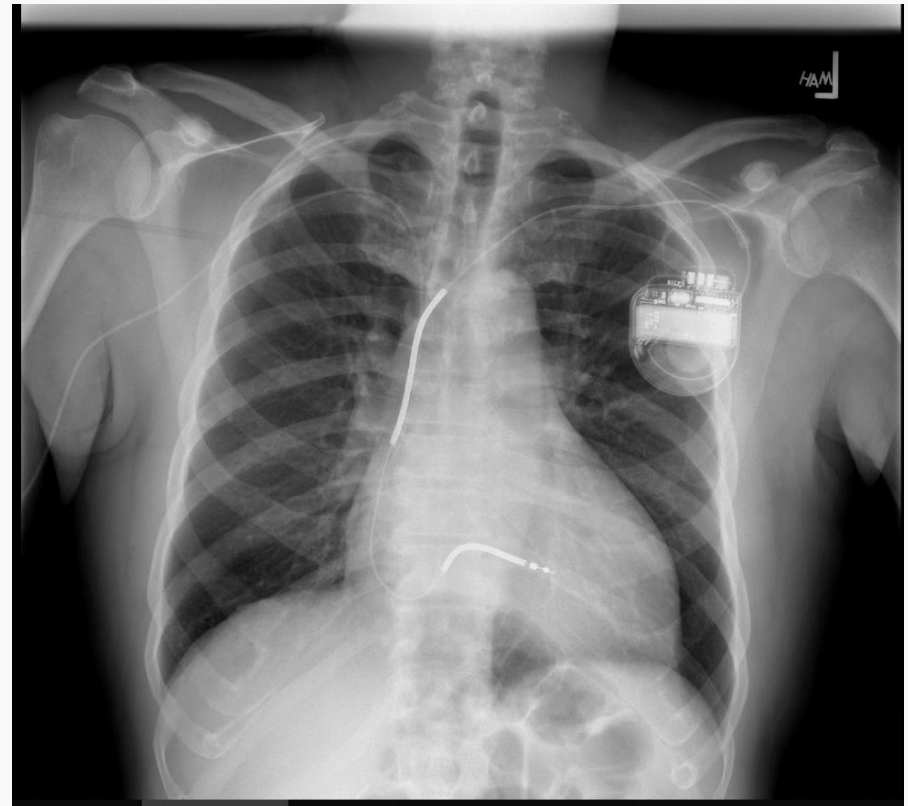
metal

Whitest

**Maximum x-ray
Absorption**

(densest tissue)

*fluid can be pus, blood or
simple fluid



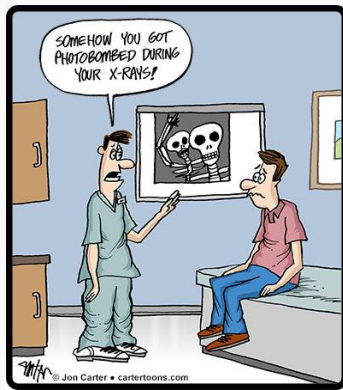


HAM

STITUTE

nic

- Basic Physics/ Radiation principles
- All of that physics basically creates the x-ray result which is essentially a shadow of the tissues that absorbed the radiation



- Understanding the radiograph

Tissue

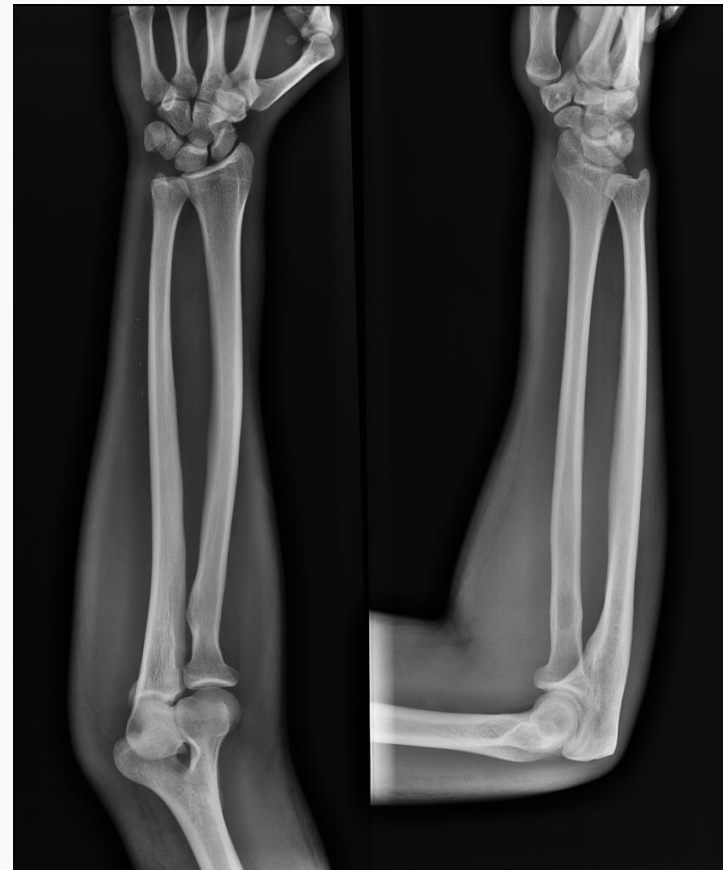
contrast
(i.e their
densities)



Image
contrast

- Allows sharp margins between structures
- If not enough contrast (diff. in density) between adjacent tissues, no sharp margins between structures on the image and the objects will blend together- **Silhouette sign****

- Understanding the radiograph
- For example: Can you identify the flexor muscles of the forearm on this radiograph?

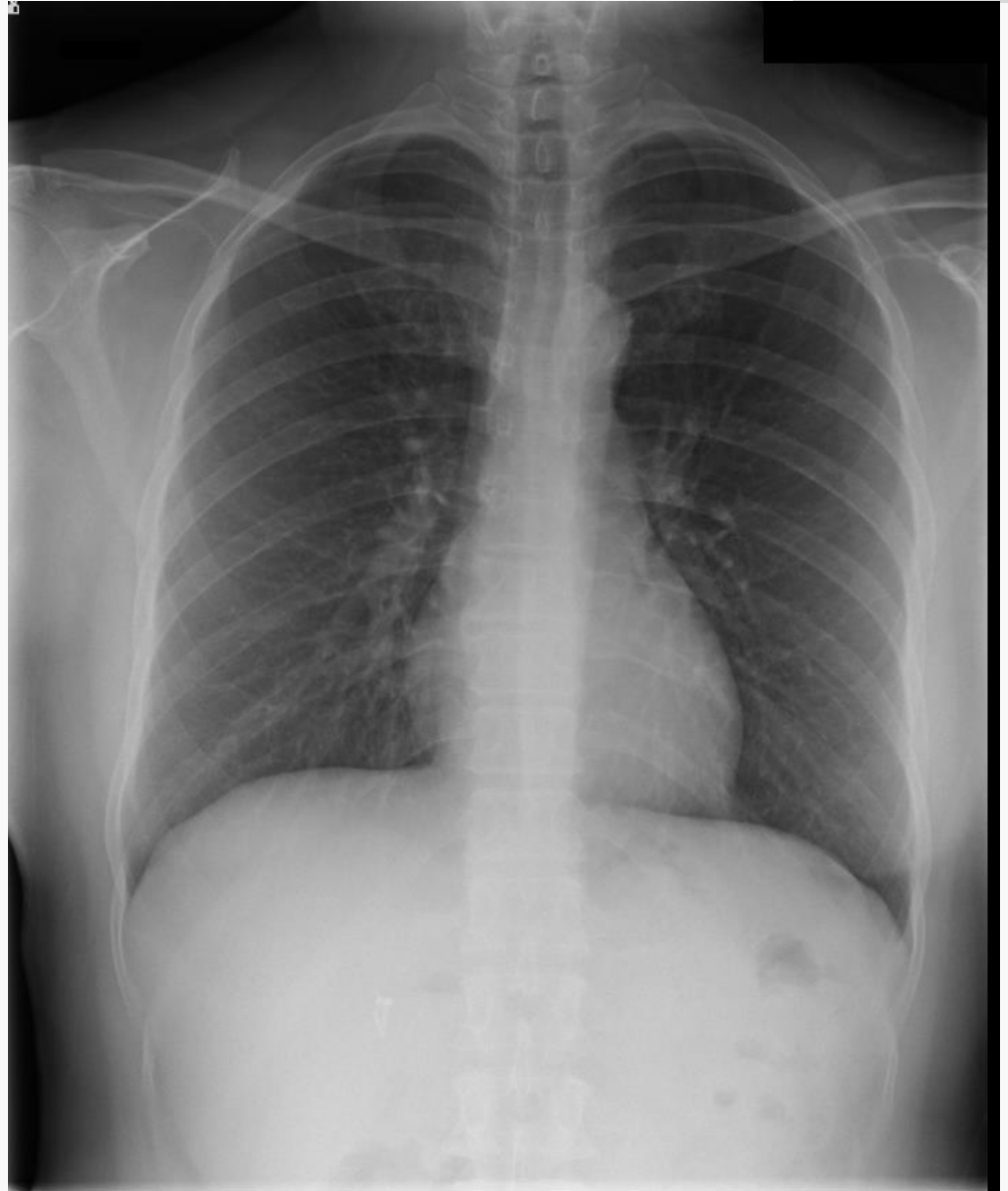


- Understanding the radiograph
- We cannot because all the muscles have the same “density” on xray....but we can see where the muscles start and the fat ends



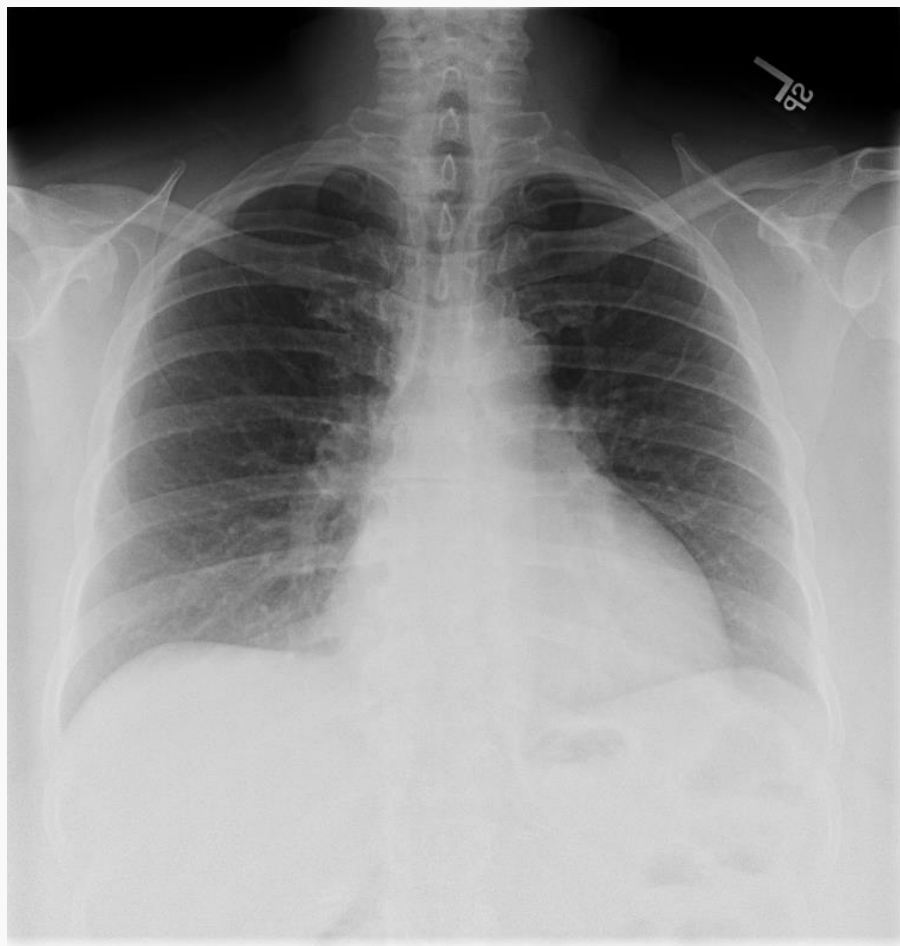
• Basic Physics/ Radiation principles

- A structure is *only* rendered visible on a radiograph by the juxtaposition of two different densities.
- When objects of the same density are next to each other they will be silhouetted
- For example: liver and diaphragm

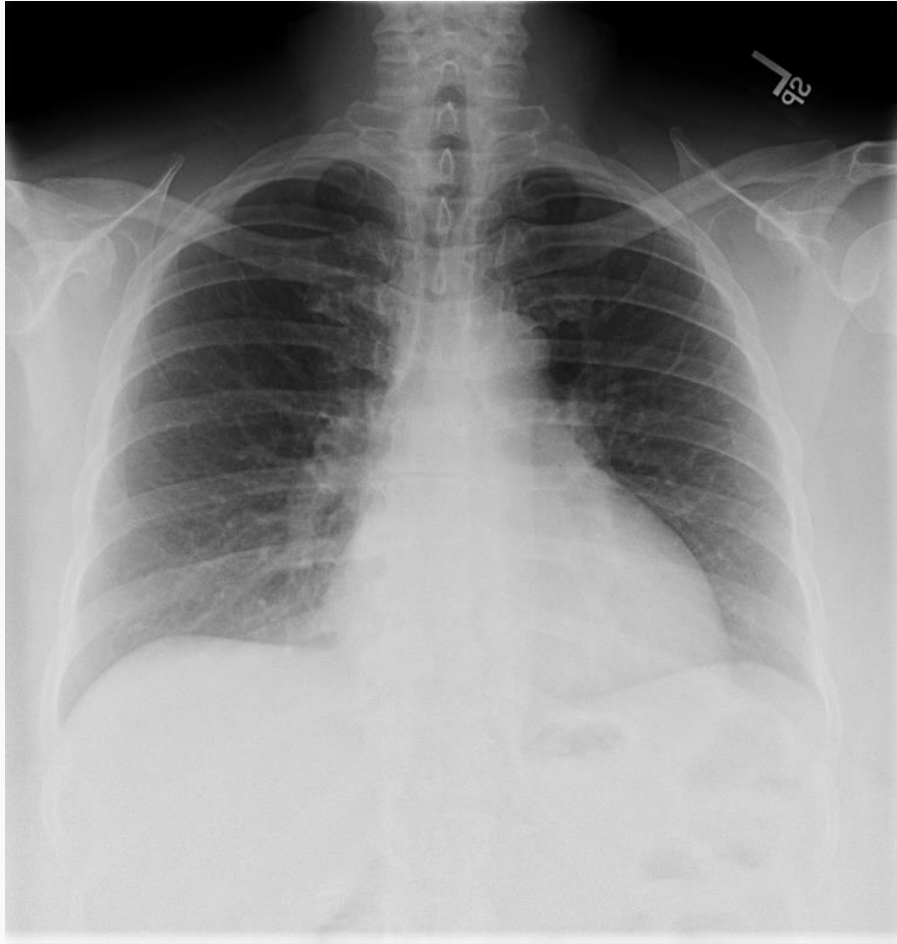


- This is a key to understanding interpretation of the chest radiograph



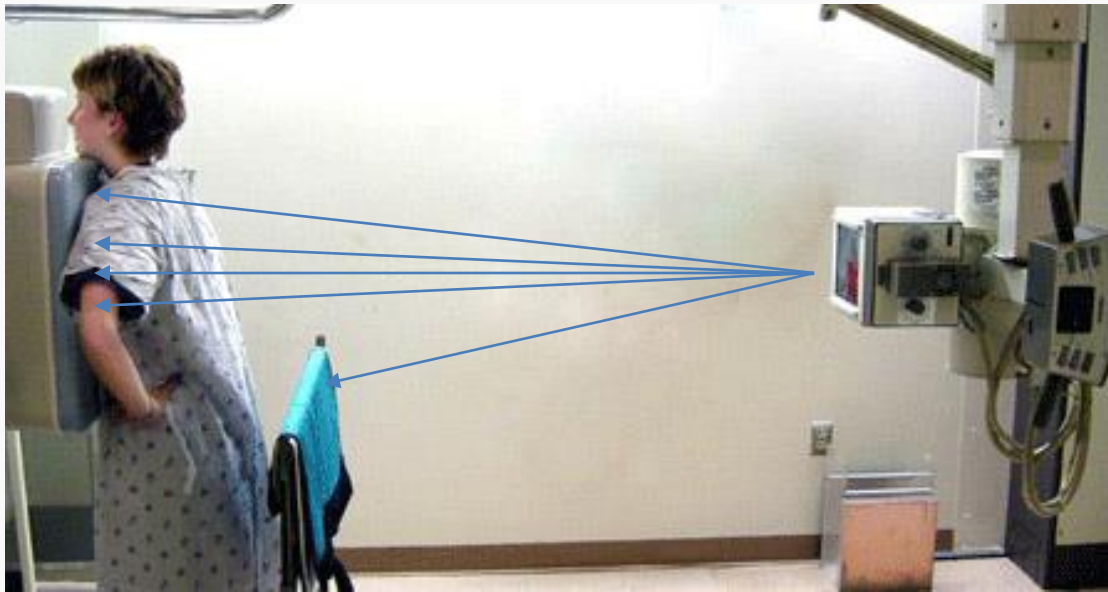


**Silhouette Sign



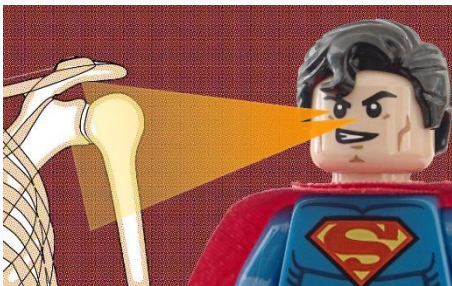
Loss of the expected interface normally created by juxtaposition of two structures of different density
No boundary can be seen between two structures of similar density

- Projection – PA or AP
 - How to tell → medial border of scapula (rule of thumb but not absolute)
- Inspiratory effort
 - 9-10 posterior ribs
- Penetration
 - thoracic intervertebral disc space just visible
- Positioning/rotation
 - medial clavicle heads equidistant to spinous process

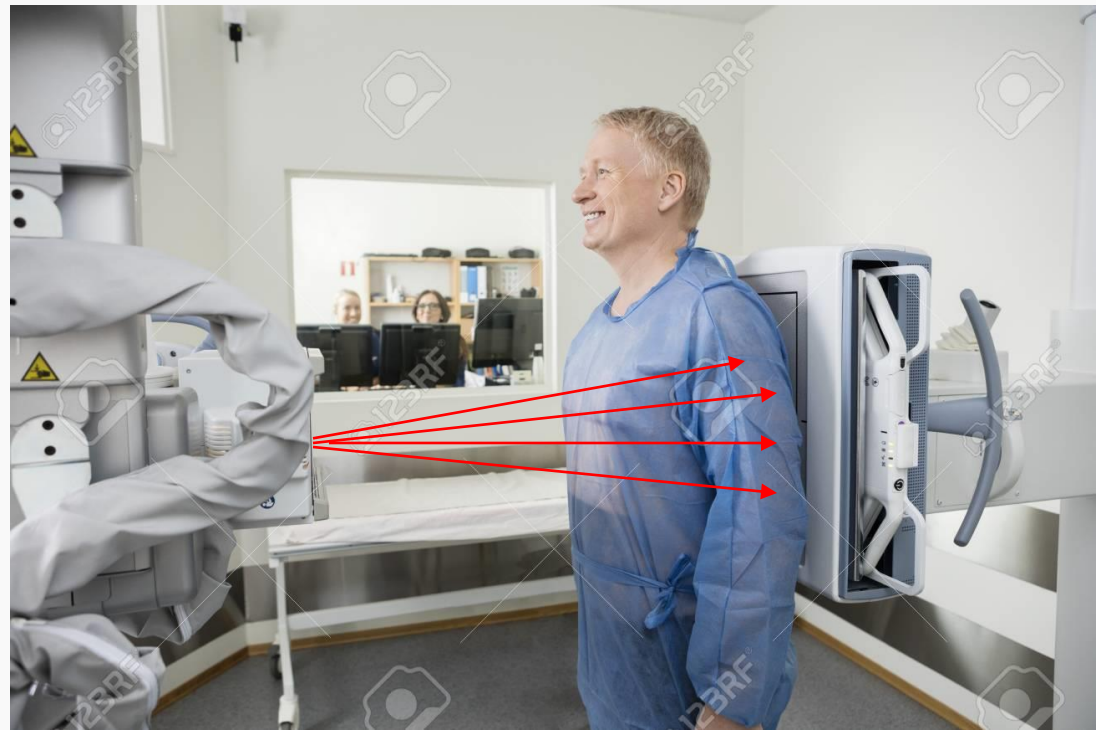


Basic X-ray unit

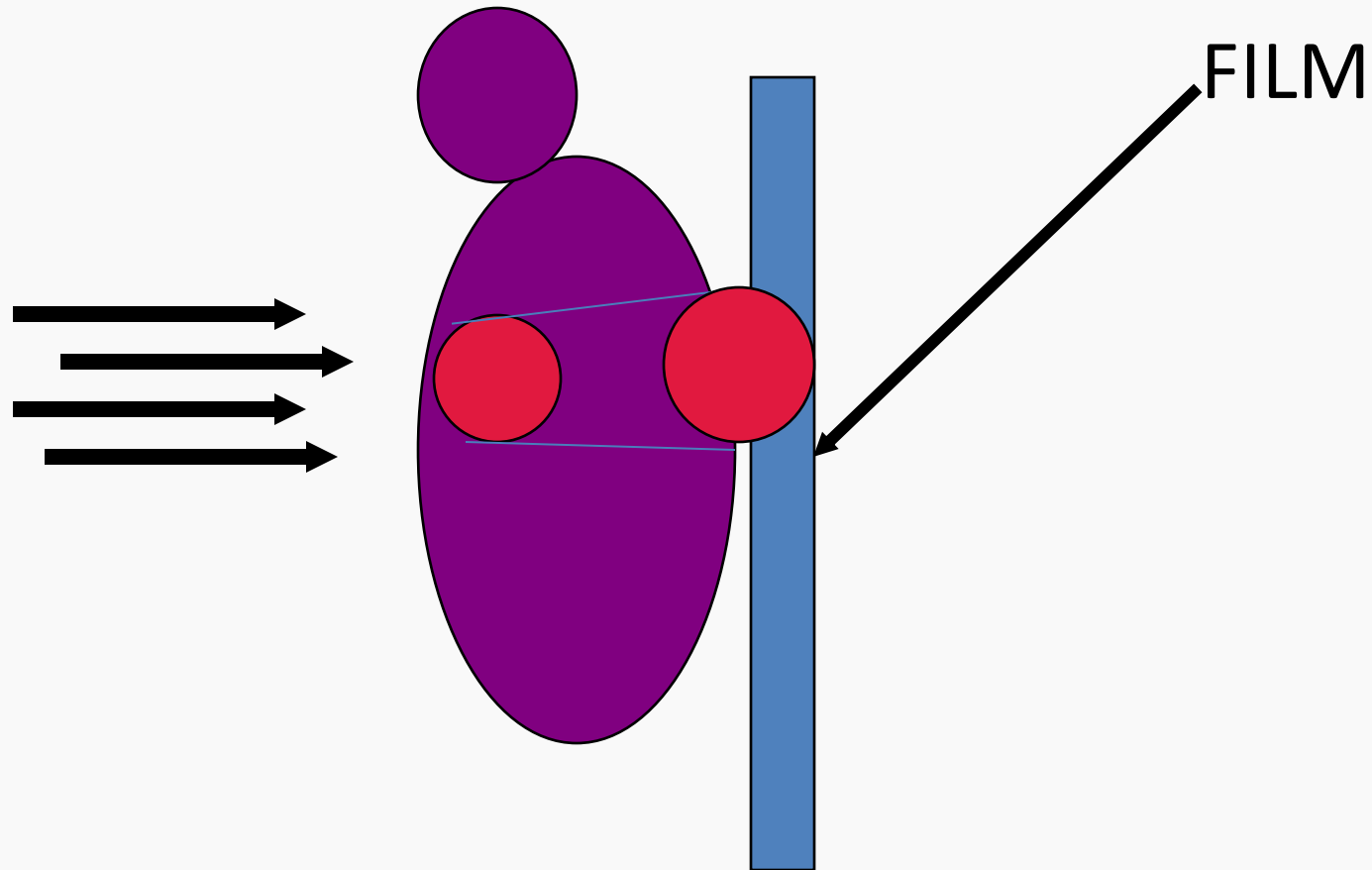
PA film



- AP film

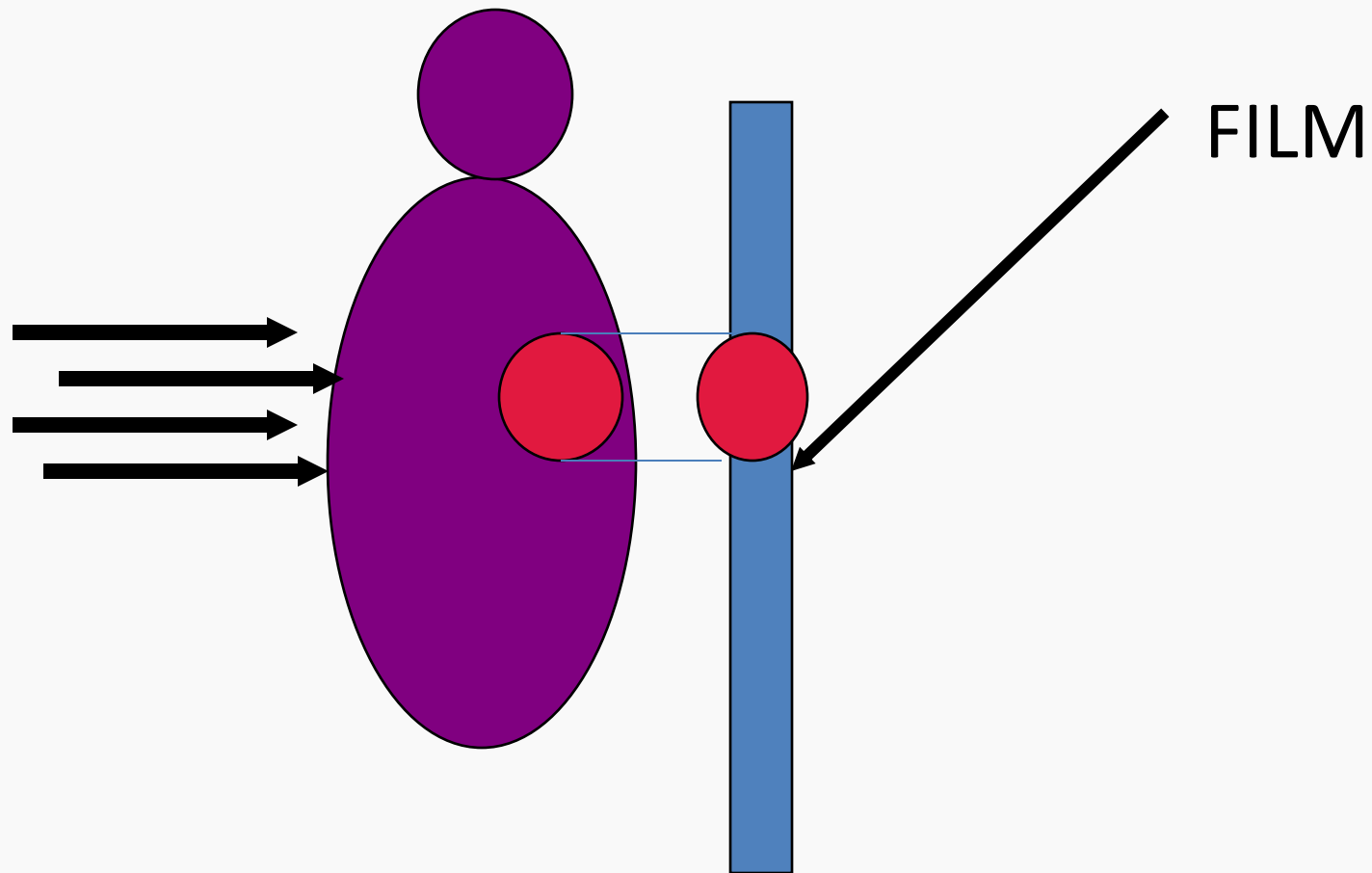


Portable (**AP** or Antero-posterior)



More magnification on AP films

PA (Posteroanterior)



better film technically due to less magnification (heart is closer to the film/cassette)



PA

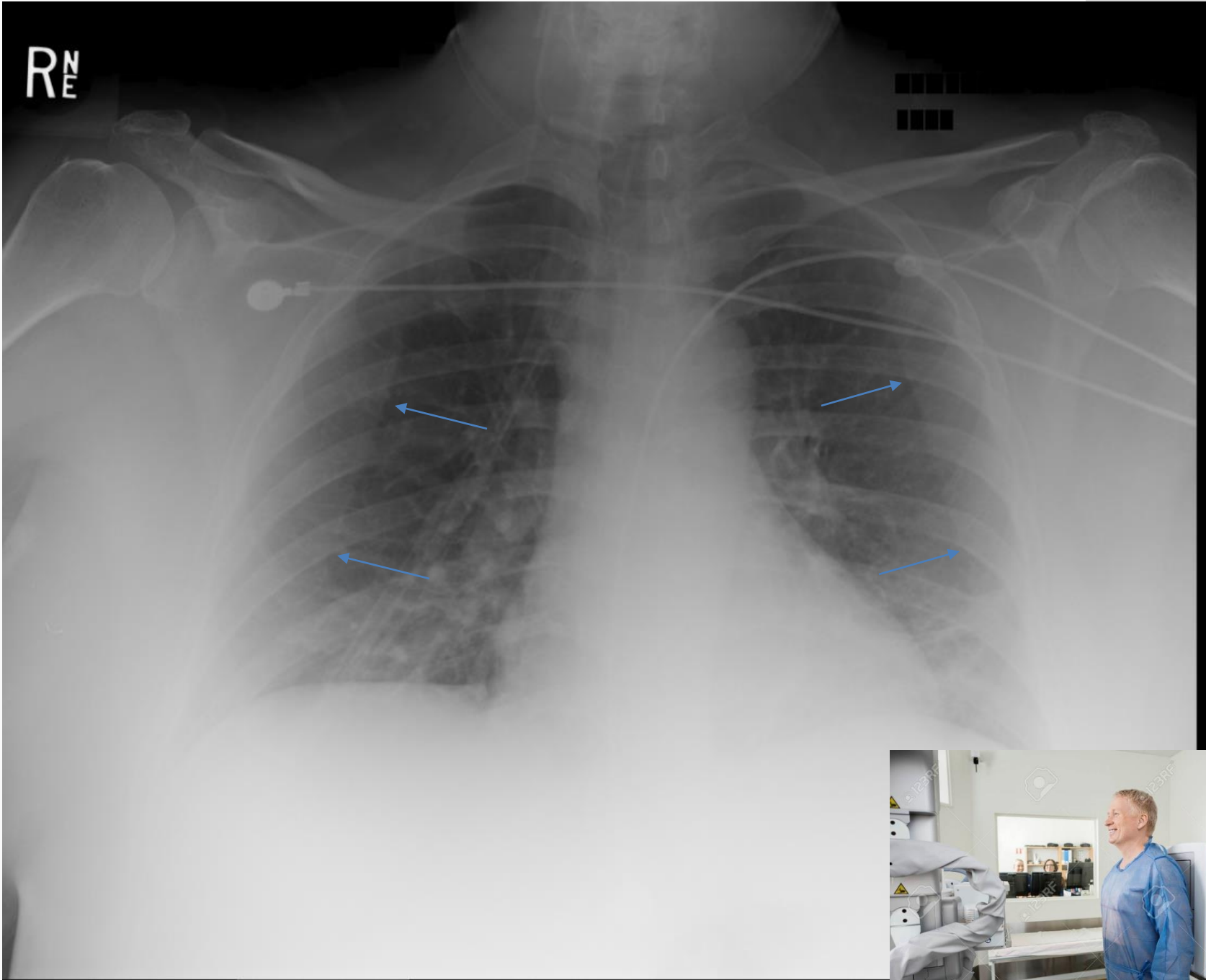


AP

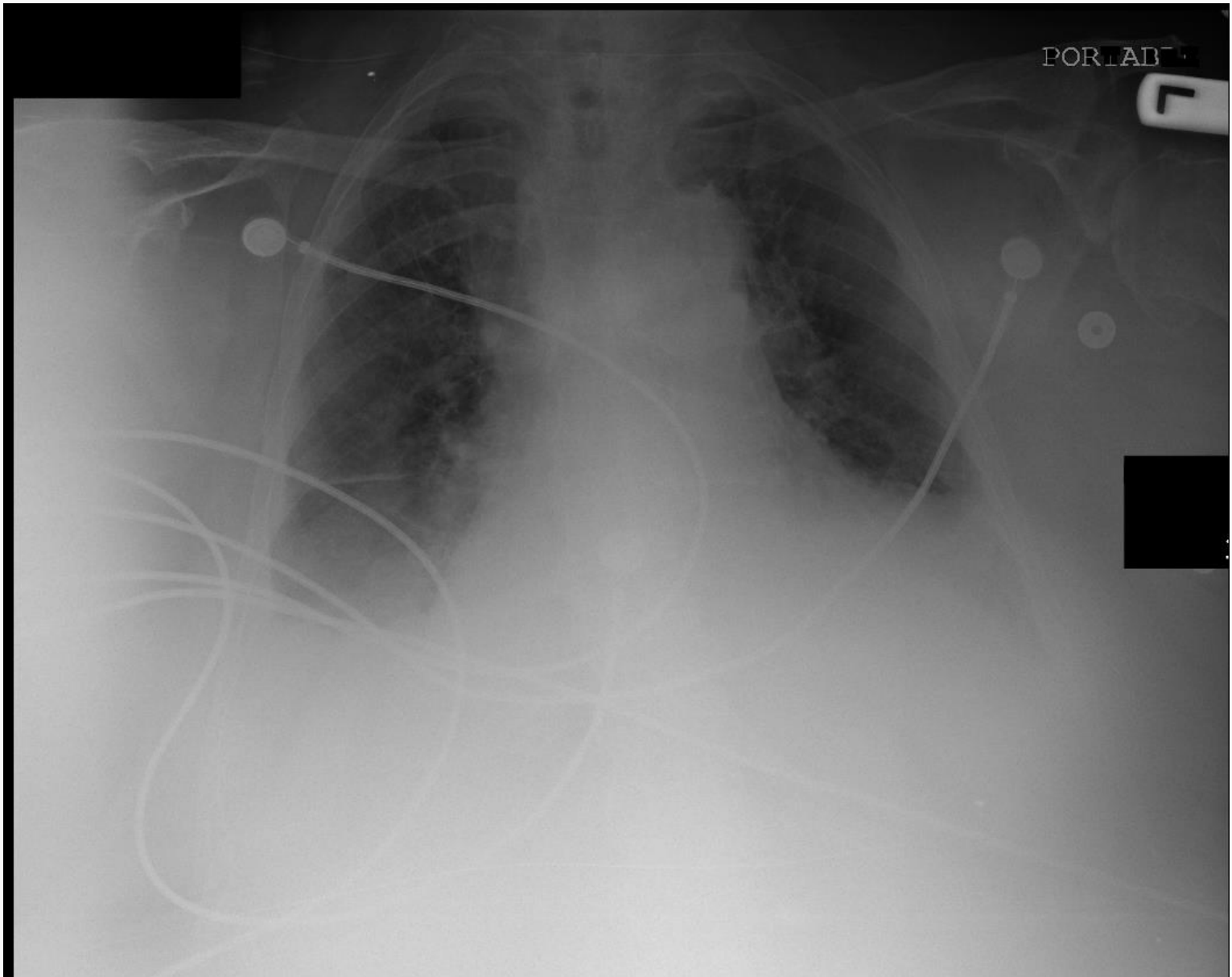
- Projection- Same patient



RN
E



- Not a steadfast rule!



- Approach to the CXR: Technical Aspects

- Projection – PA or AP
- Inspiratory effort
 - 9-10 posterior ribs
- Penetration
 - thoracic intervertebral disc space just visible
- Positioning/rotation
 - medial clavicle heads equidistant to spinous process



PA



AP

- Low lung volumes- Same patient

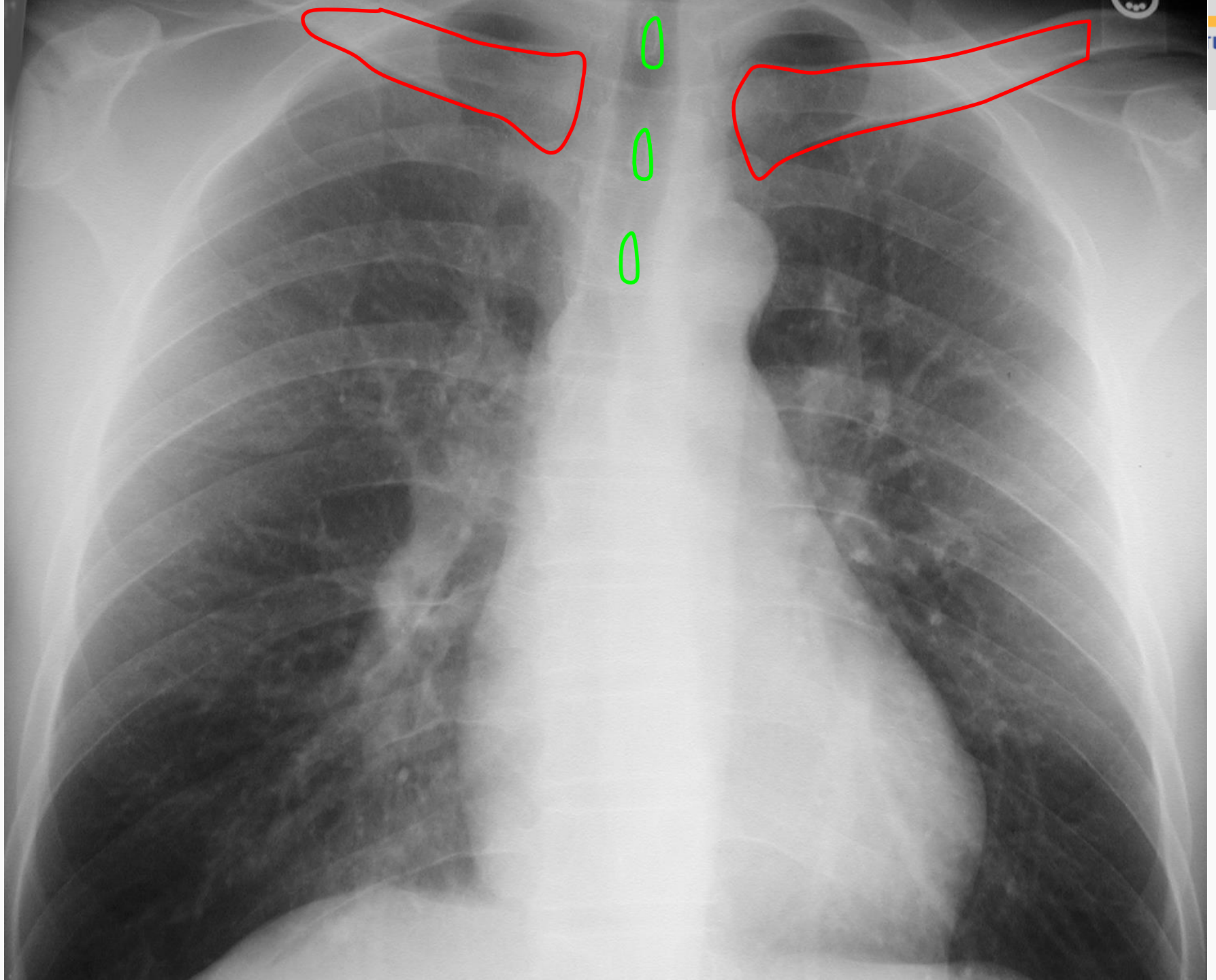
- Approach to the CXR: Technical Aspects

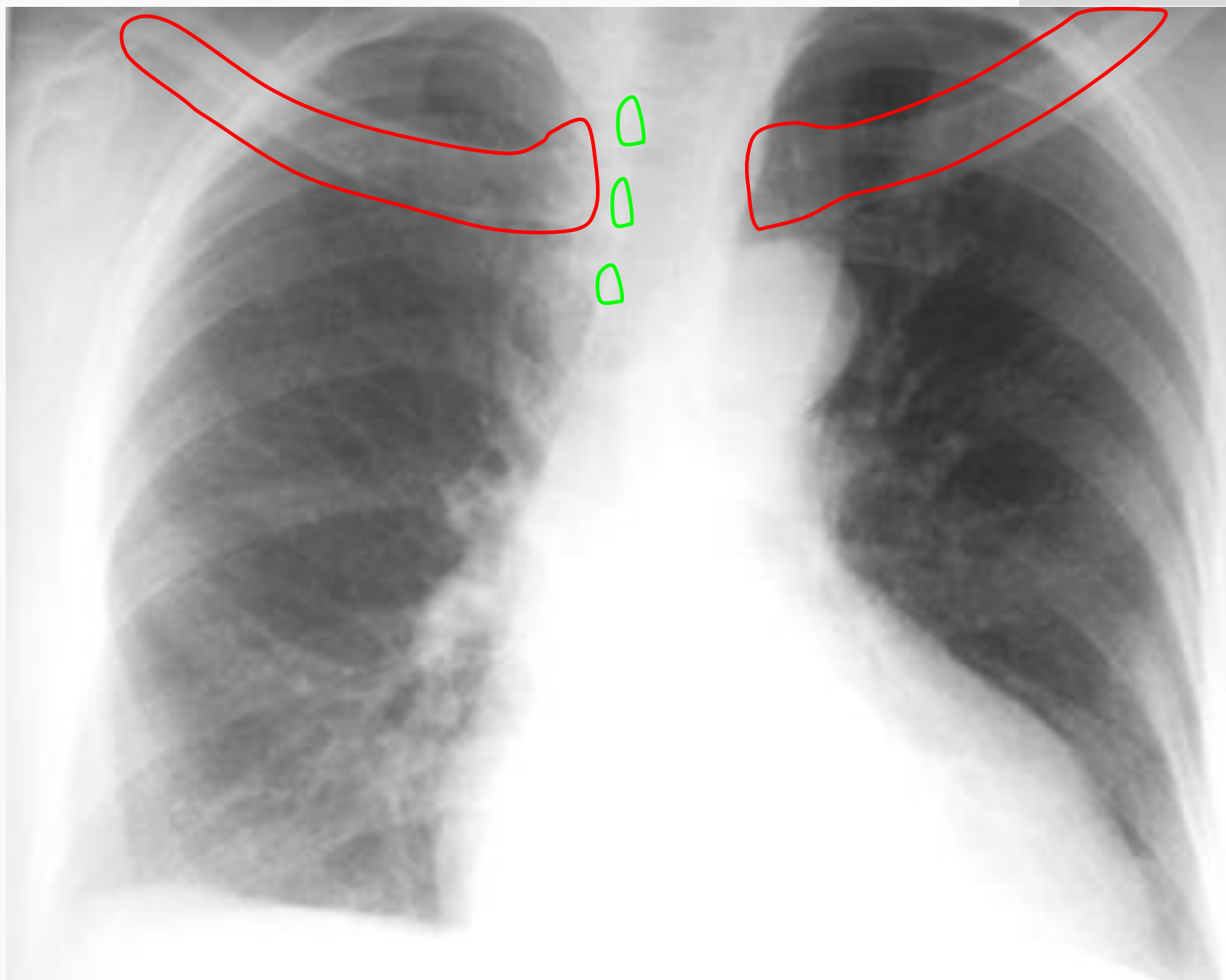
- Projection – PA or AP
- Inspiratory effort
 - 9-10 posterior ribs
- Penetration
 - thoracic intervertebral disc space just visible
- Positioning/rotation
 - medial clavicle heads equidistant to spinous process





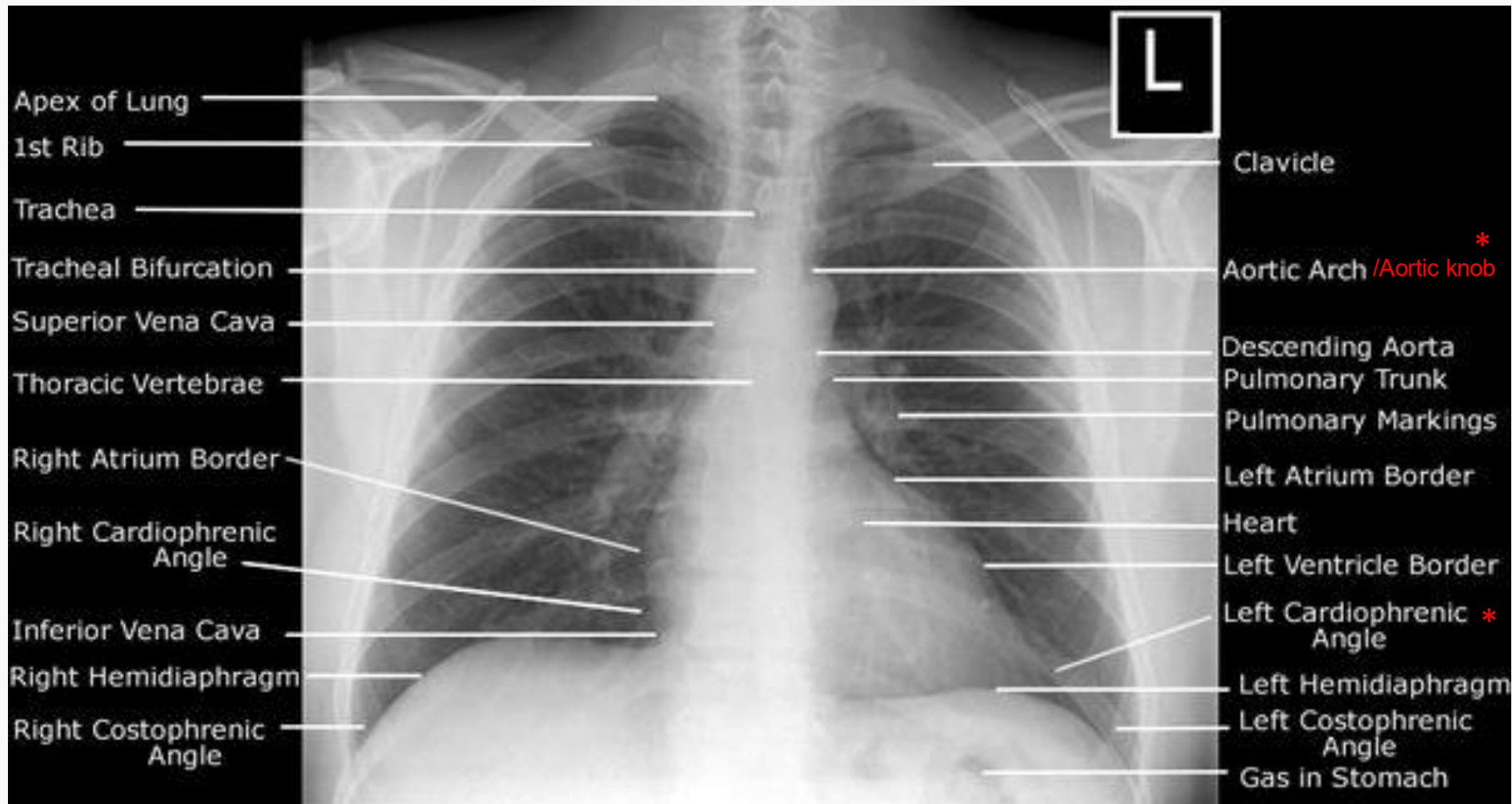
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- Projection – PA or AP
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- Penetration
 - thoracic intervertebral disc space just visible
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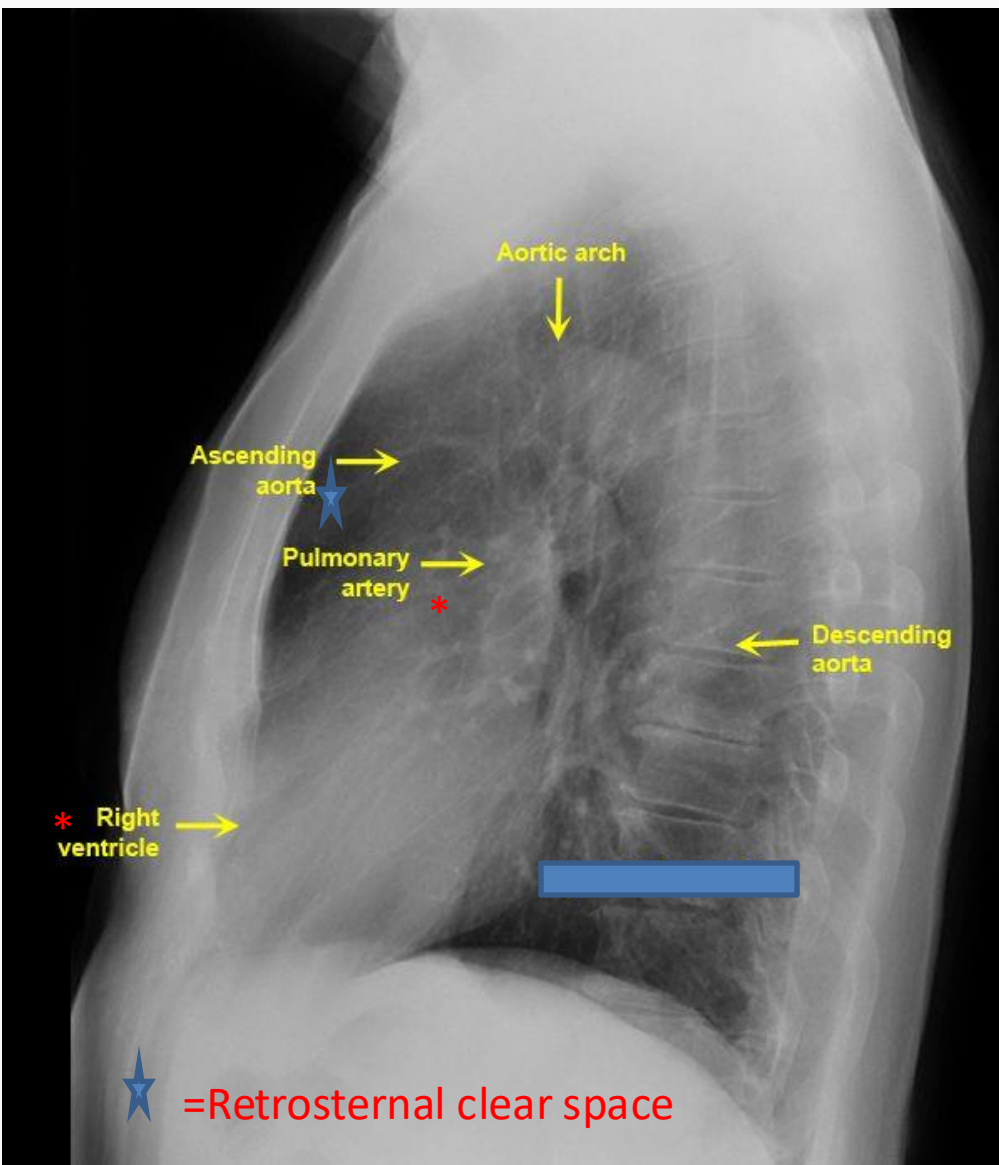


Anatomy

• Normal Frontal Chest X-ray

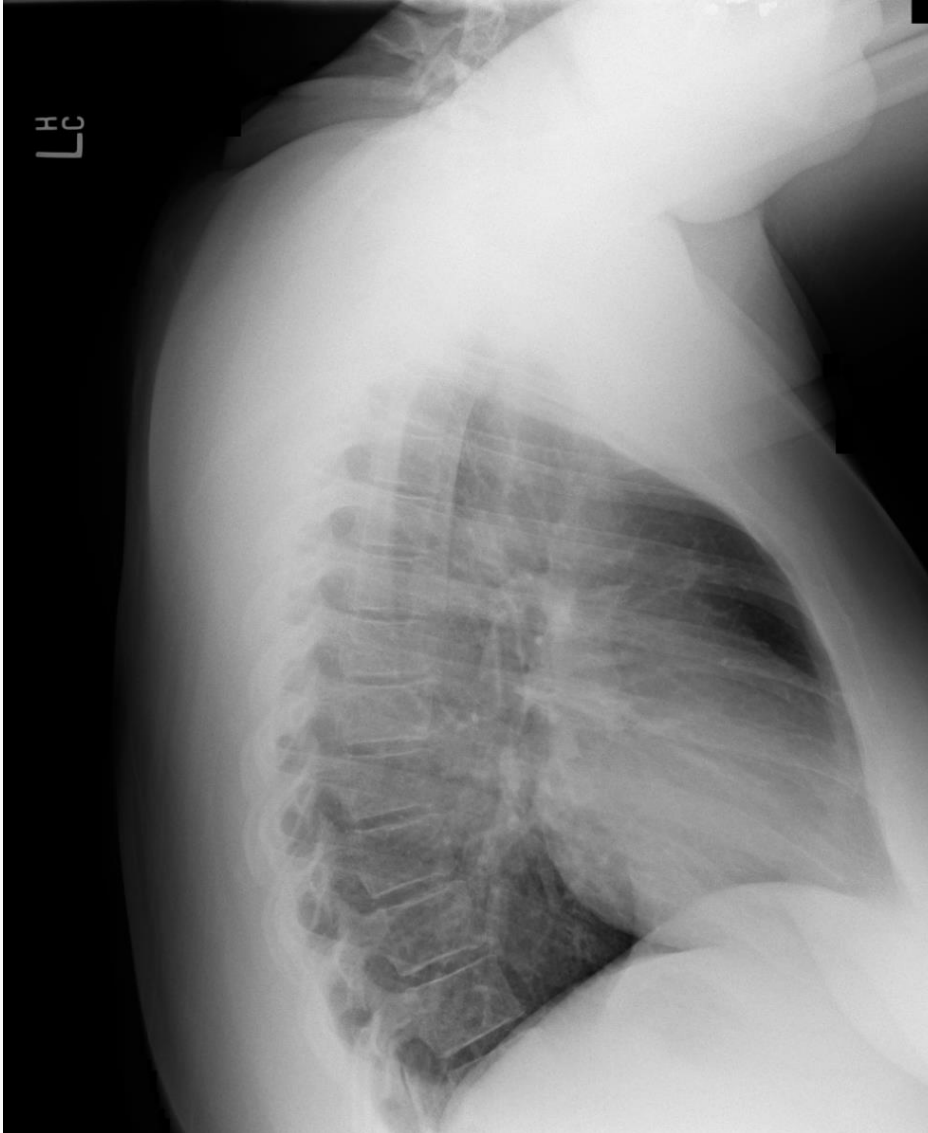


- Left heart border = left ventricle
- Right heart border= right atrium





Who do you identify left and right
Diaphragms?



Identify left and
right
Diaphragm

The right
hemidiaphragm
will be seen
in its entirety
while the left will
not be seen well
where the heart
sits on top of it

Because of the
....?

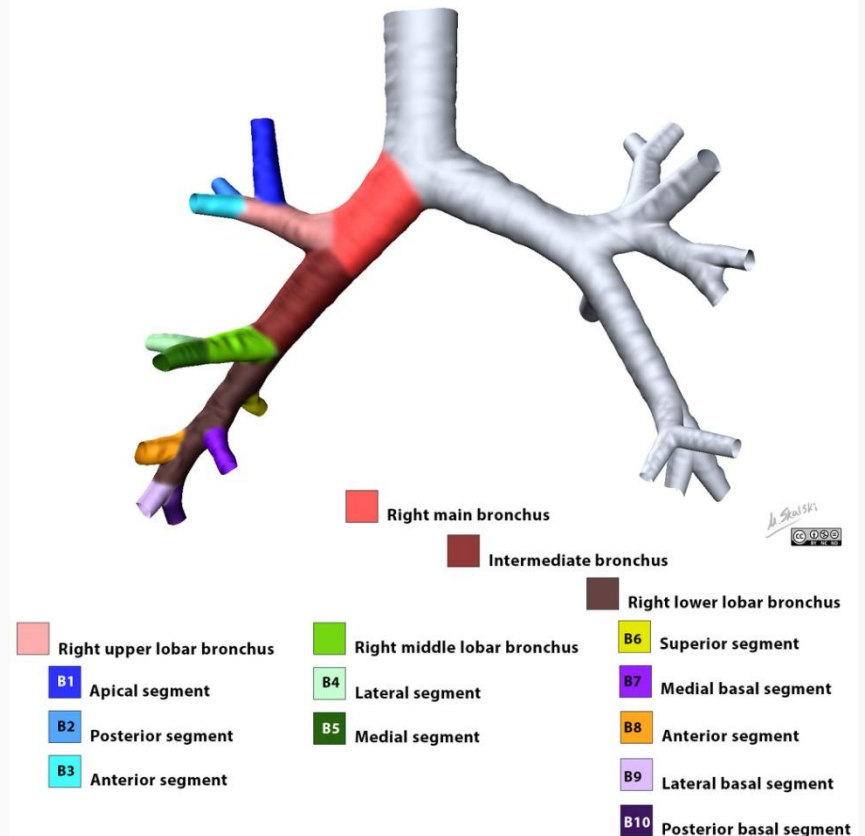
Bronchopulmonary anatomy



Right lung Normal Anatomy

- The right lung is subdivided into three lobes with ten segments.
 - Right upper lobe bronchus lobe
 - apical segment
 - posterior segment
 - anterior segment
 - Right middle lobe bronchus-intermedius – right middle lobe (or more correctly - just middle lobe)
 - lateral segment
 - medial segment
 - Right lower lobe bronchus
 - superior segment
 - medial segment
 - anterior segment
 - lateral segment
 - posterior segment

- Useful mnemonic to remember the bronchopulmonary segments are:



A PALM

Seed **M**akes **A**nother **L**ittle **P**alm (right lung)

• Left lung Normal Anatomy

- The left lung is subdivided into two lobes and thereby, into eight segments.

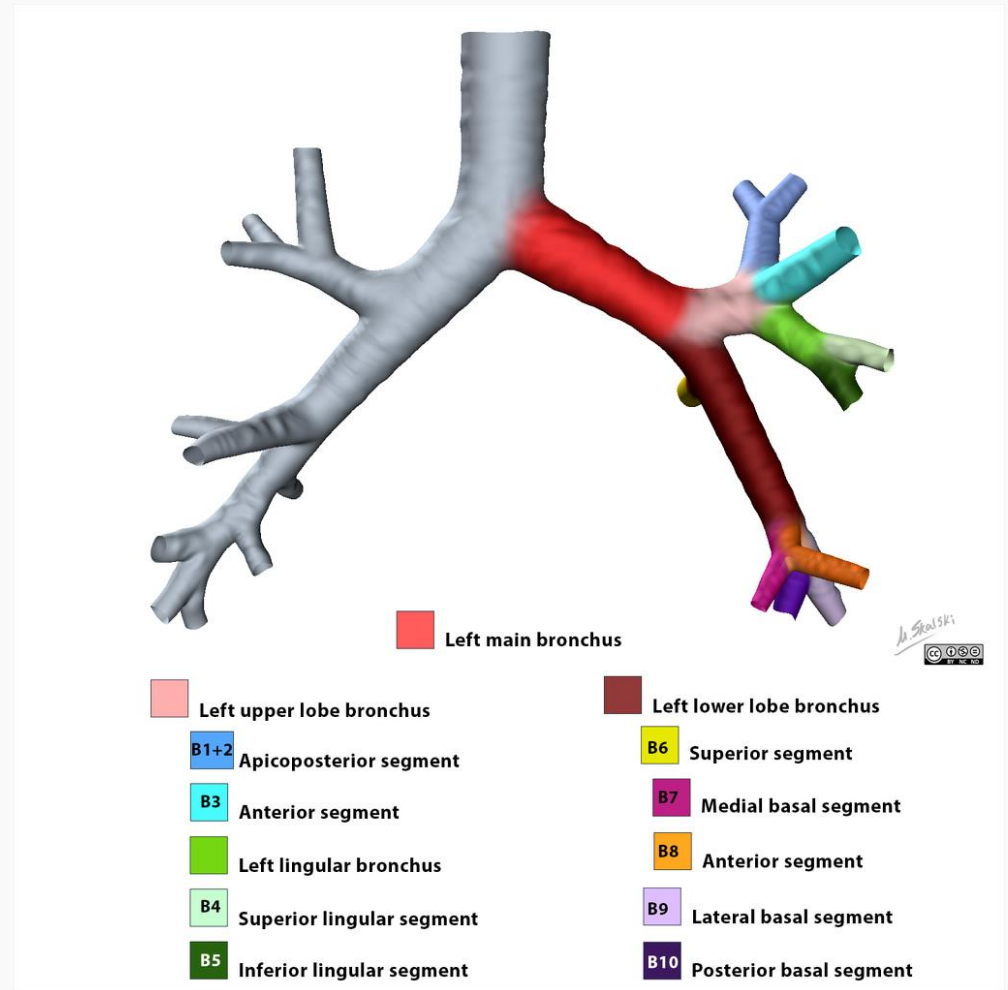
- left upper lobe bronchus

- apicoposterior segment
- Lingular lobe bronchus
 - » superior lingular segment
 - » inferior lingular segment
- anterior segment

- left lower lobe

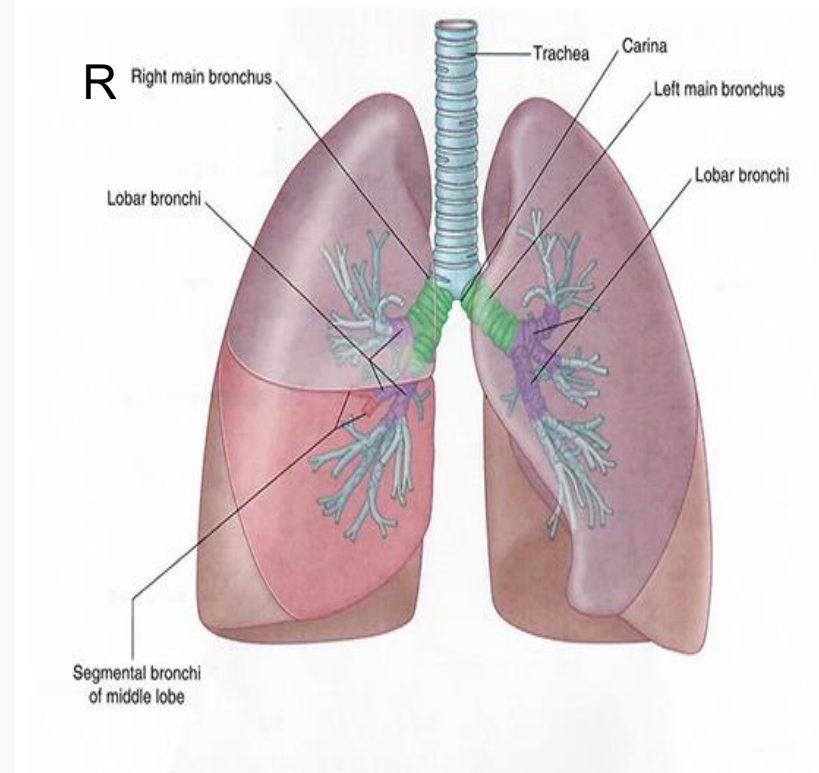
- anteromedial segment
- lateral segment
- posterior segment
- superior segment

- Useful mnemonic to remember the bronchopulmonary segments are:
 - **ASIA ALPS** (left lung)

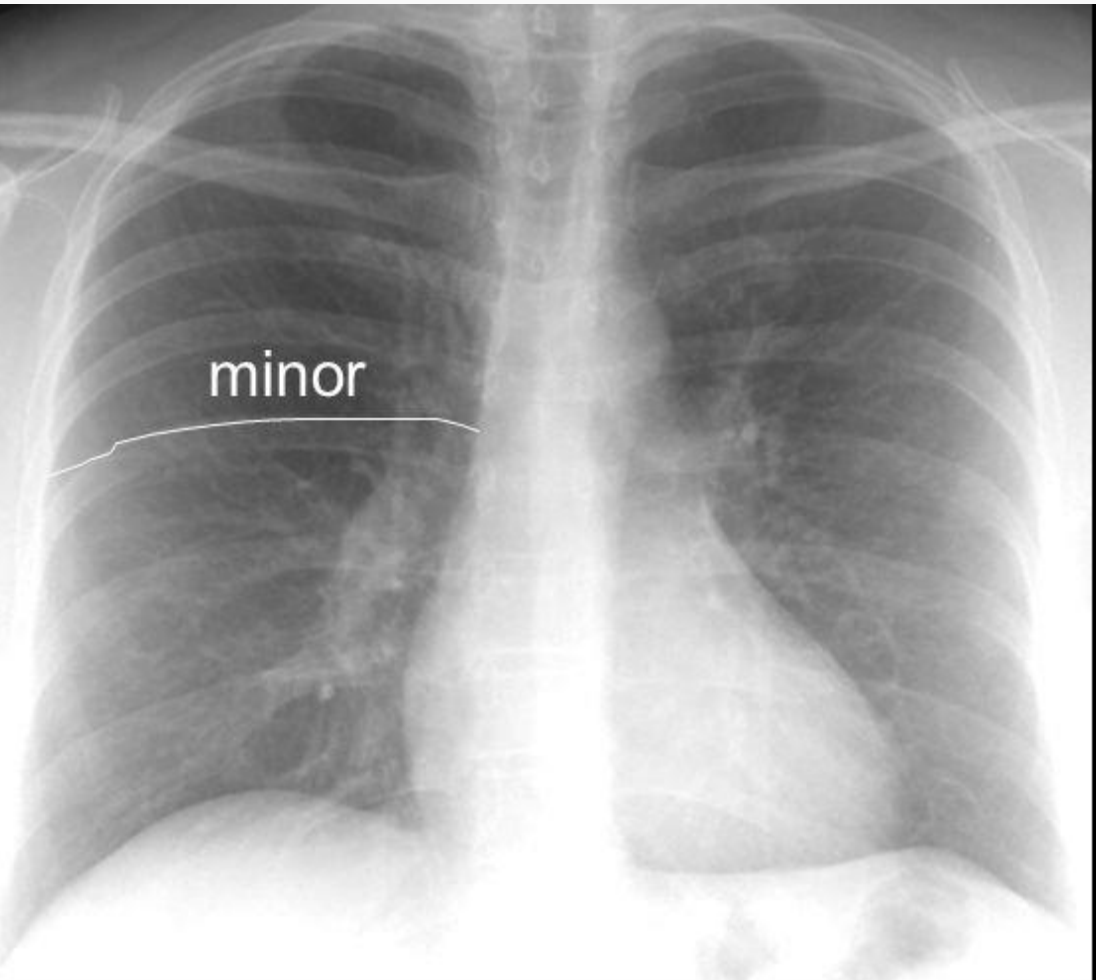


Case courtesy of Dr Matt Skalski, Radiopaedia.org. From the case rID: 23285

- Fissures
 - Major fissure(oblique)-right and left lungs
 - Minor fissure(horizontal)-right

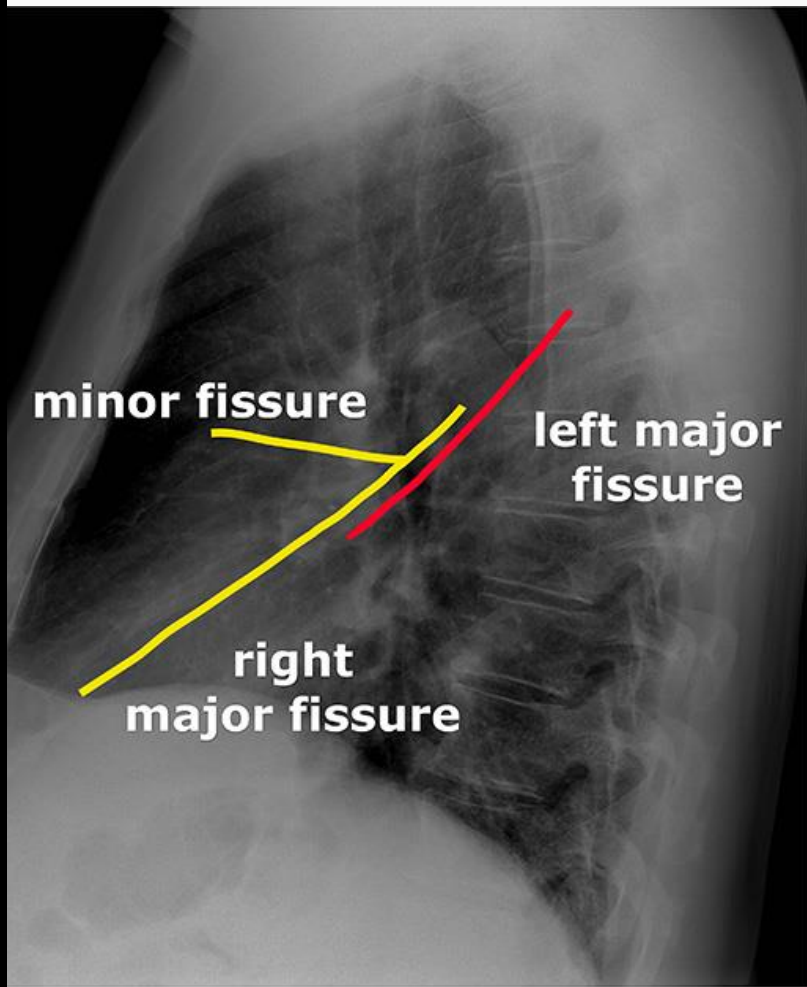


- Normal Anatomy



minor

This is an anteroposterior (AP) chest X-ray. A white line is drawn across the right lung field, representing the minor fissure. The label 'minor' is placed above this line.



minor fissure

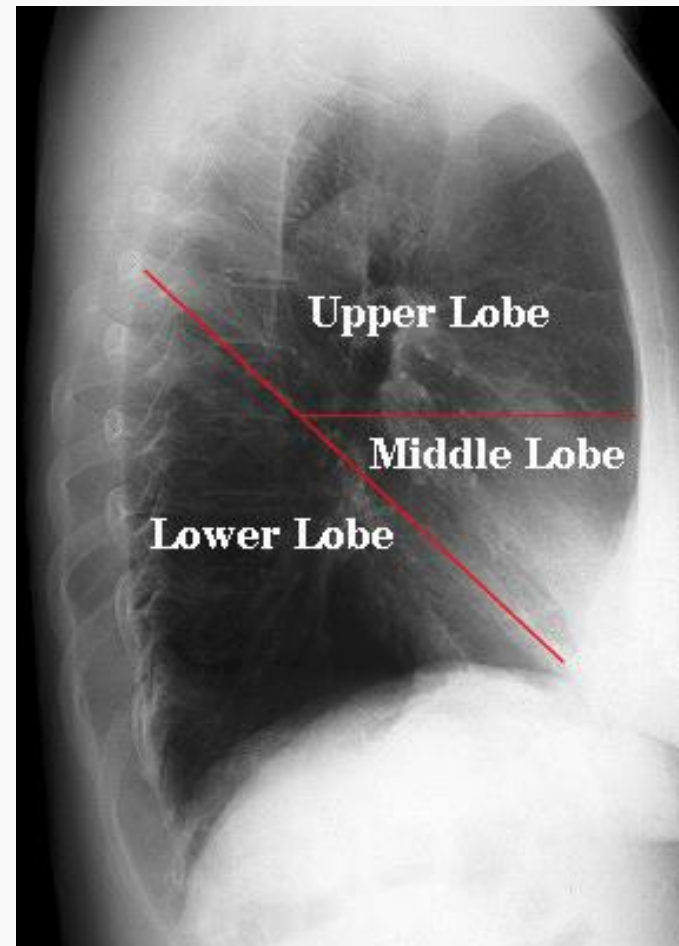
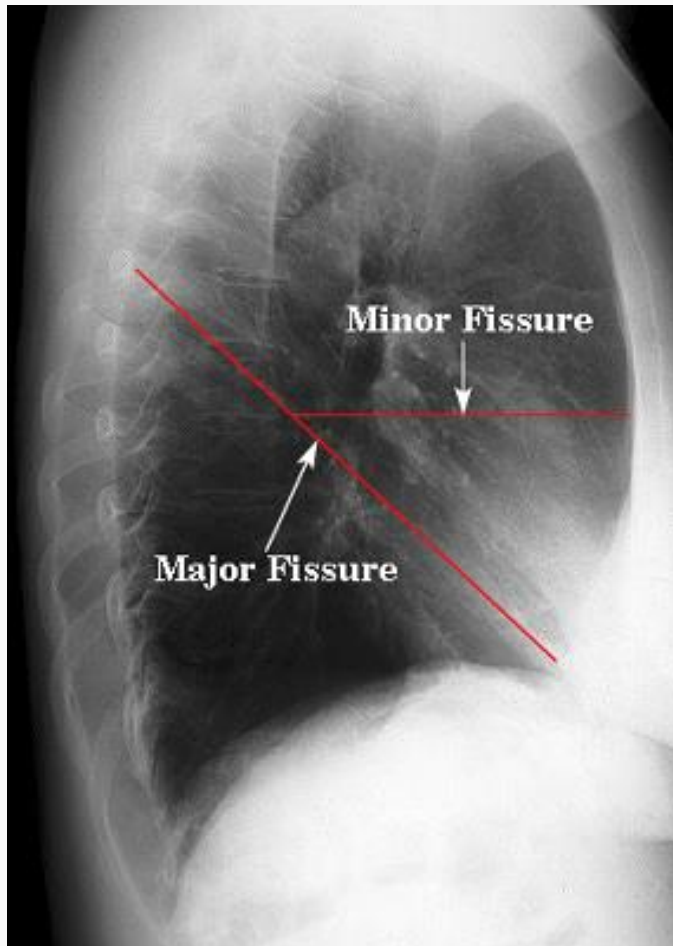
This is a lateral chest X-ray. The minor fissure is highlighted with a yellow line. The label 'minor fissure' is placed to the left of this line.

left major
fissure

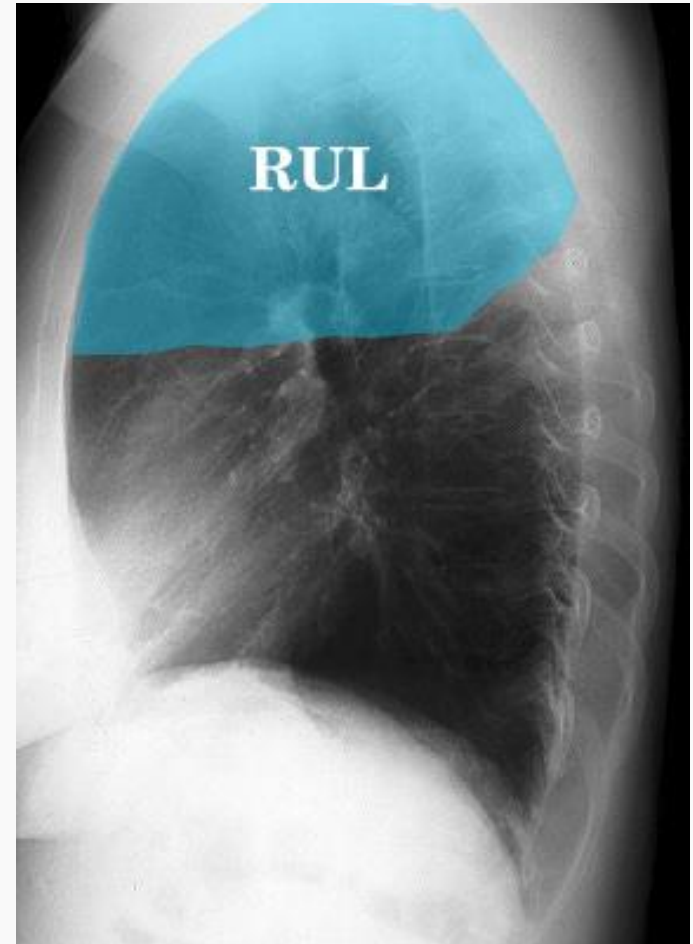
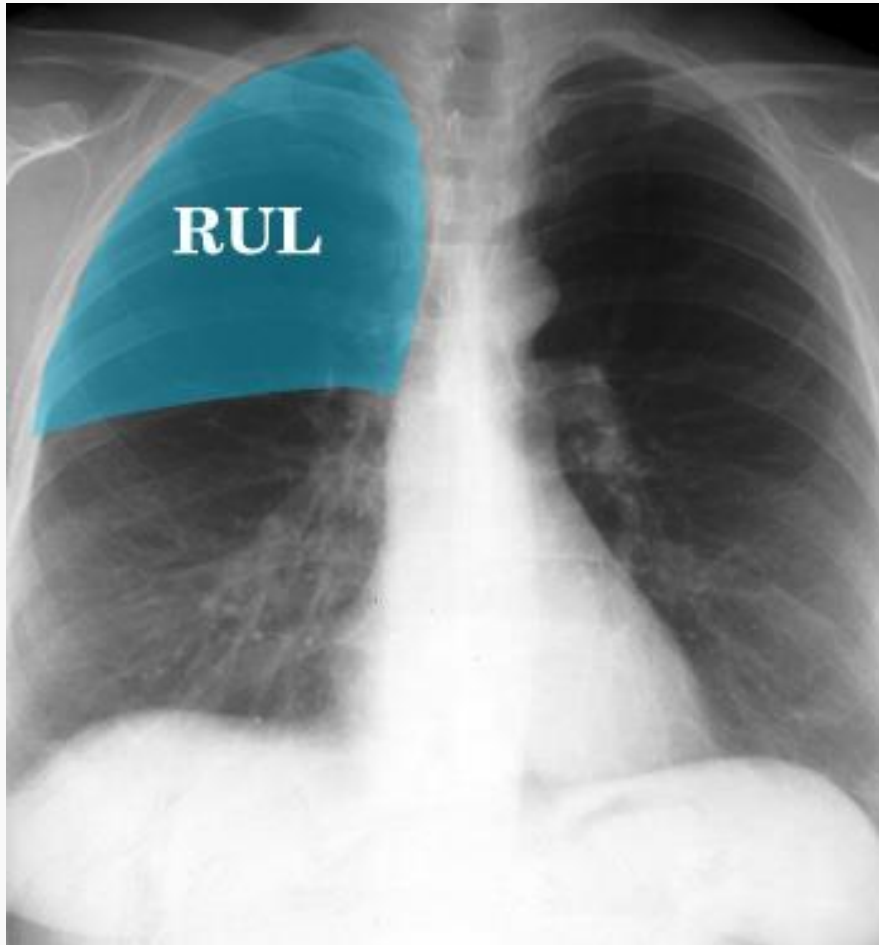
This is a lateral chest X-ray. The left major fissure is highlighted with a red line. The label 'left major fissure' is placed to the right of this line.

right
major fissure

This is a lateral chest X-ray. The right major fissure is highlighted with a yellow line. The label 'right major fissure' is placed below this line.

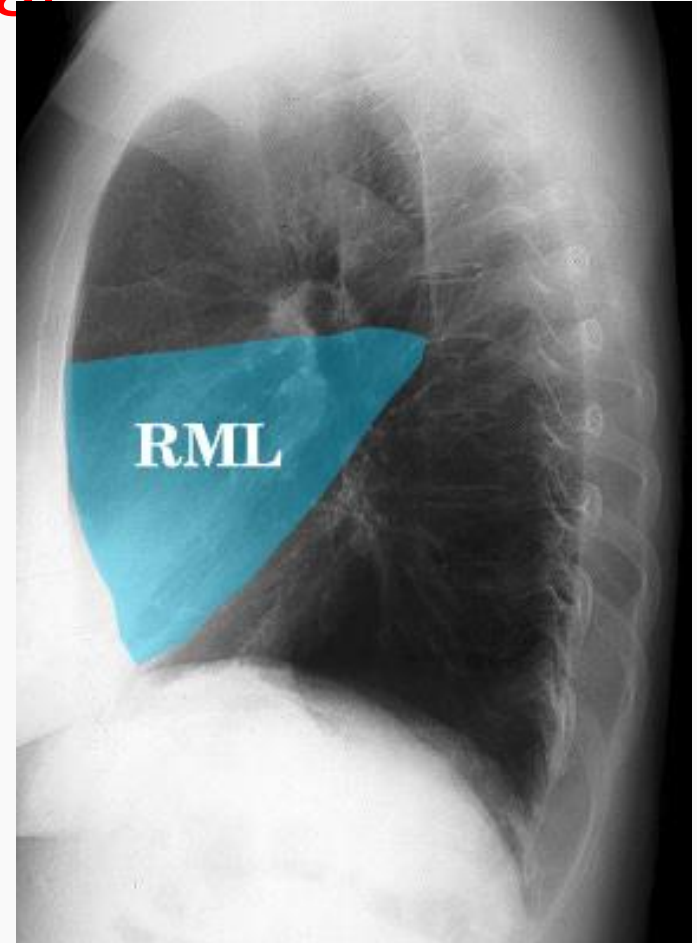
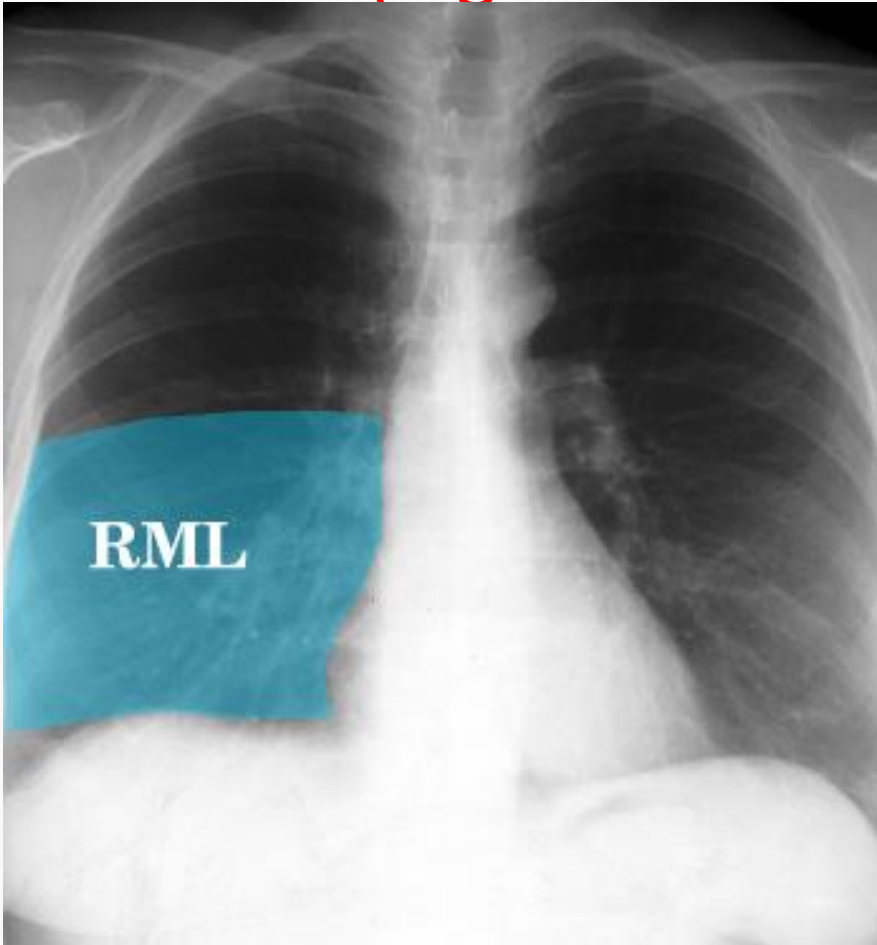


- Right Sided Fissures



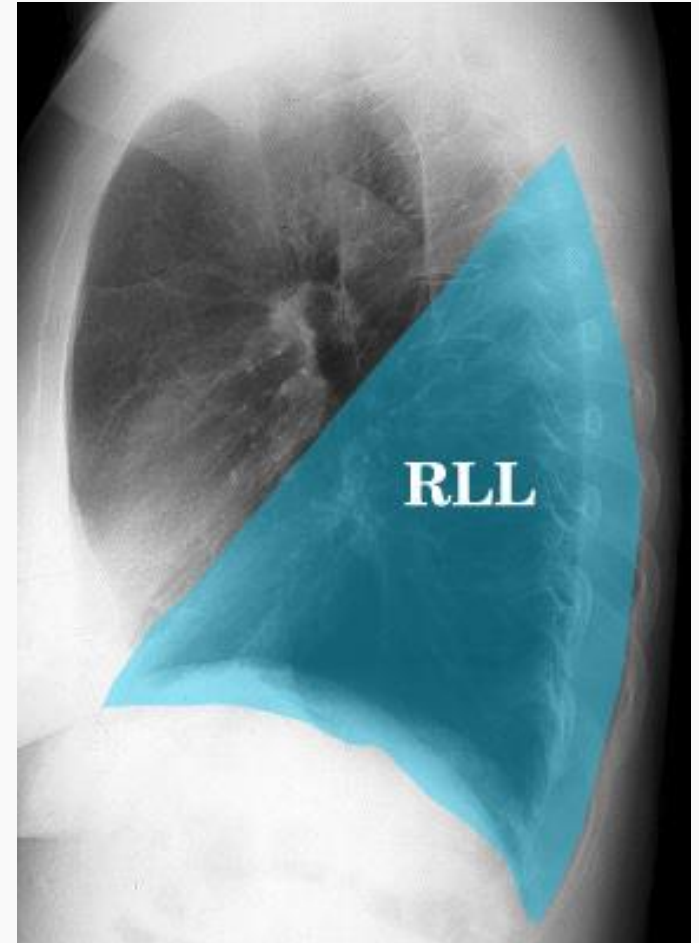
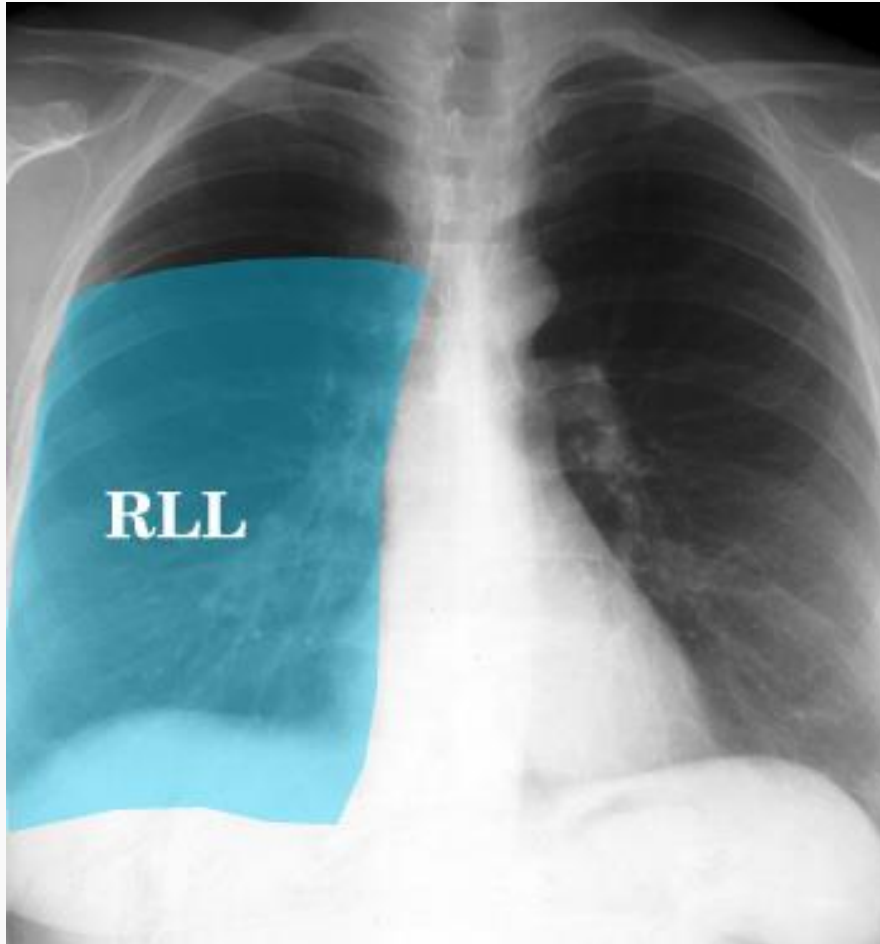
- RUL (Right Upper Lung)

- RML (Right Middle Lung)

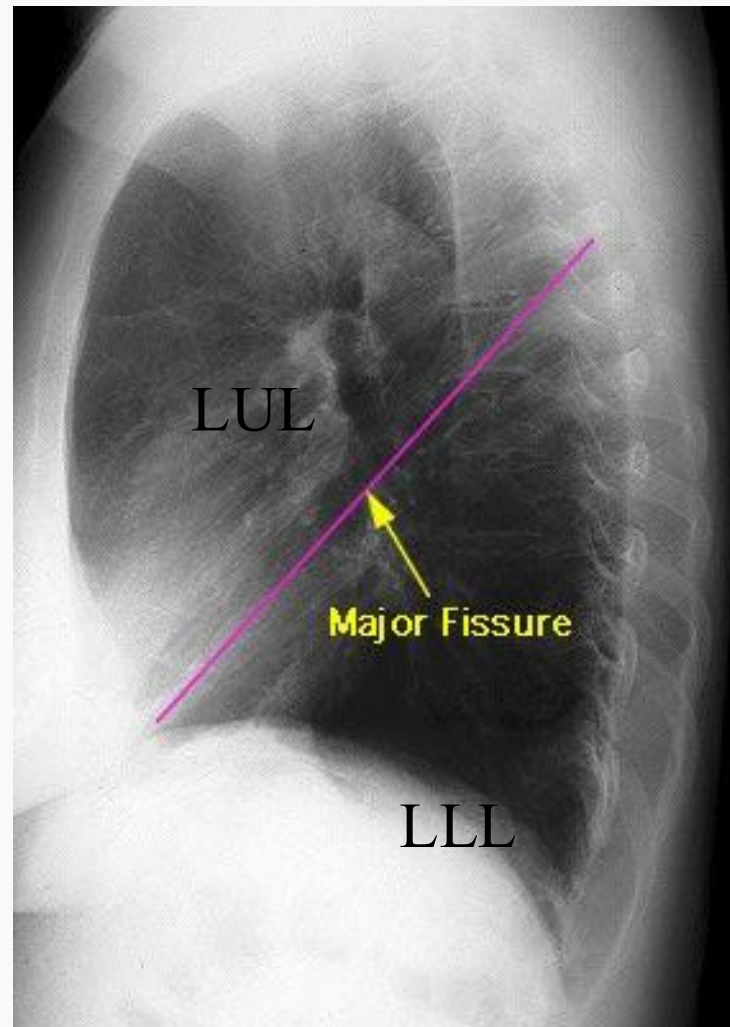


- Medial and lateral segments- The medial segment allows visualization of the right heart border

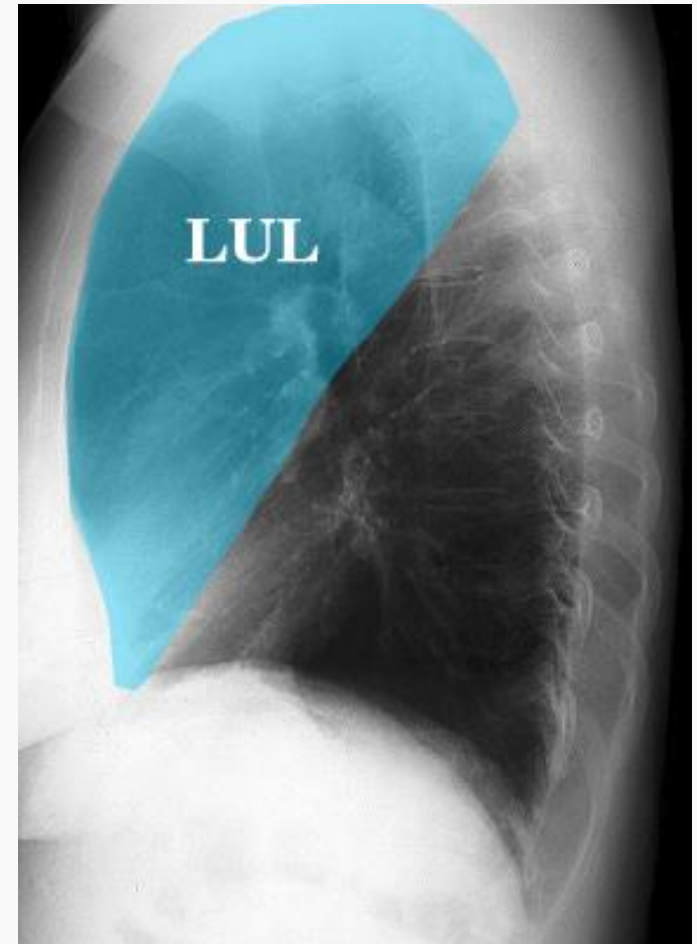
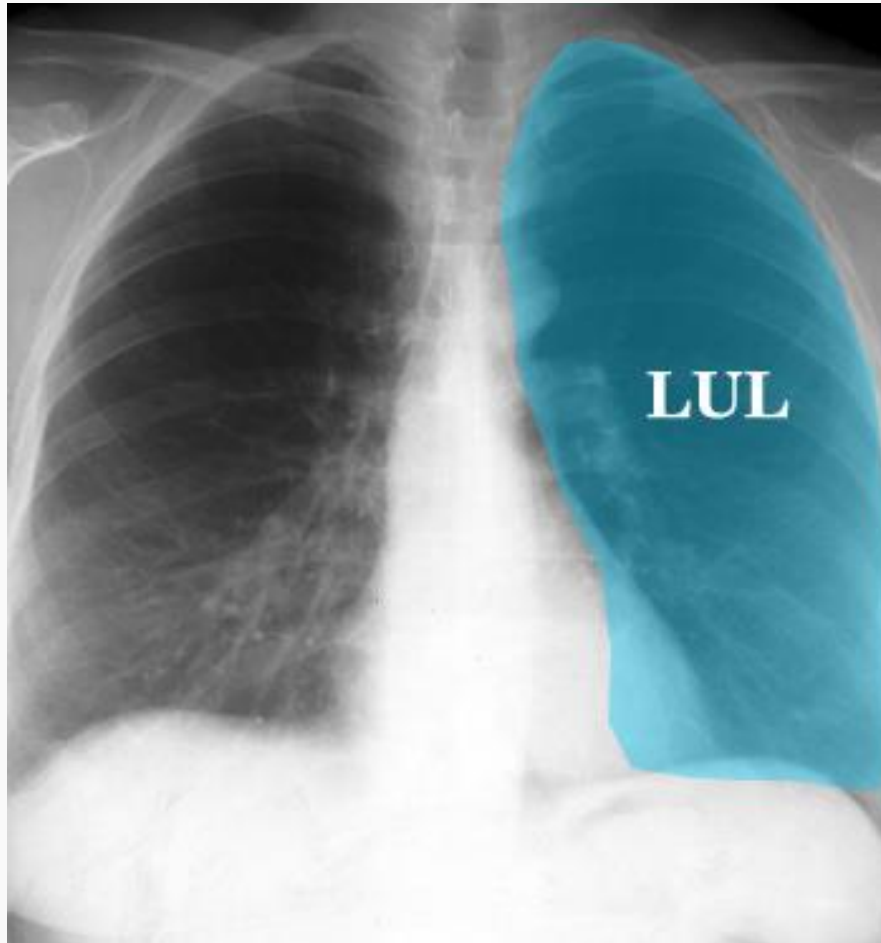
- RLL (Right Lower Lung)



- Left Side Fissure

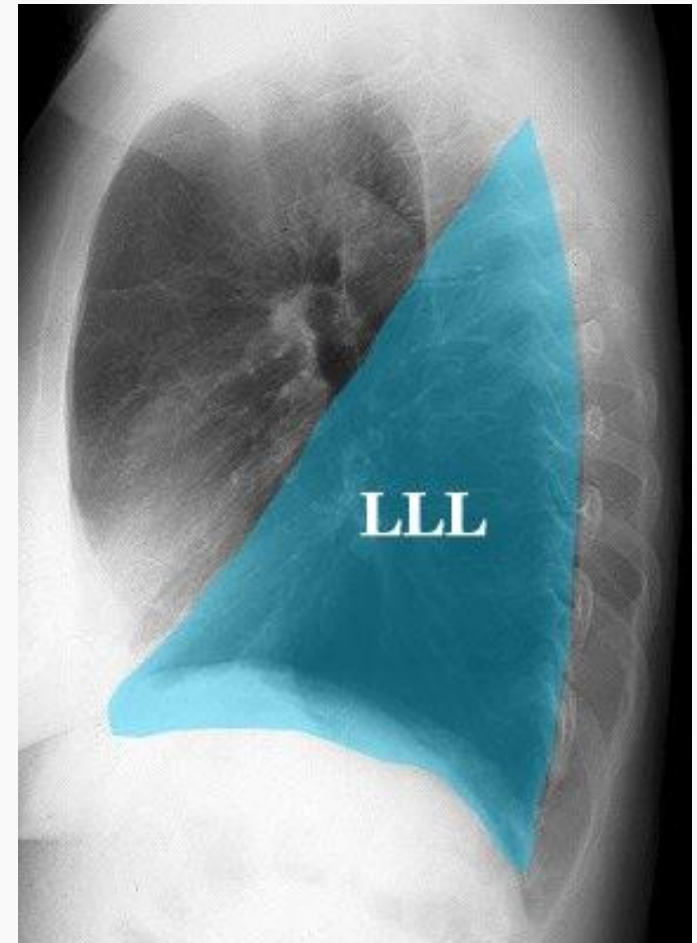
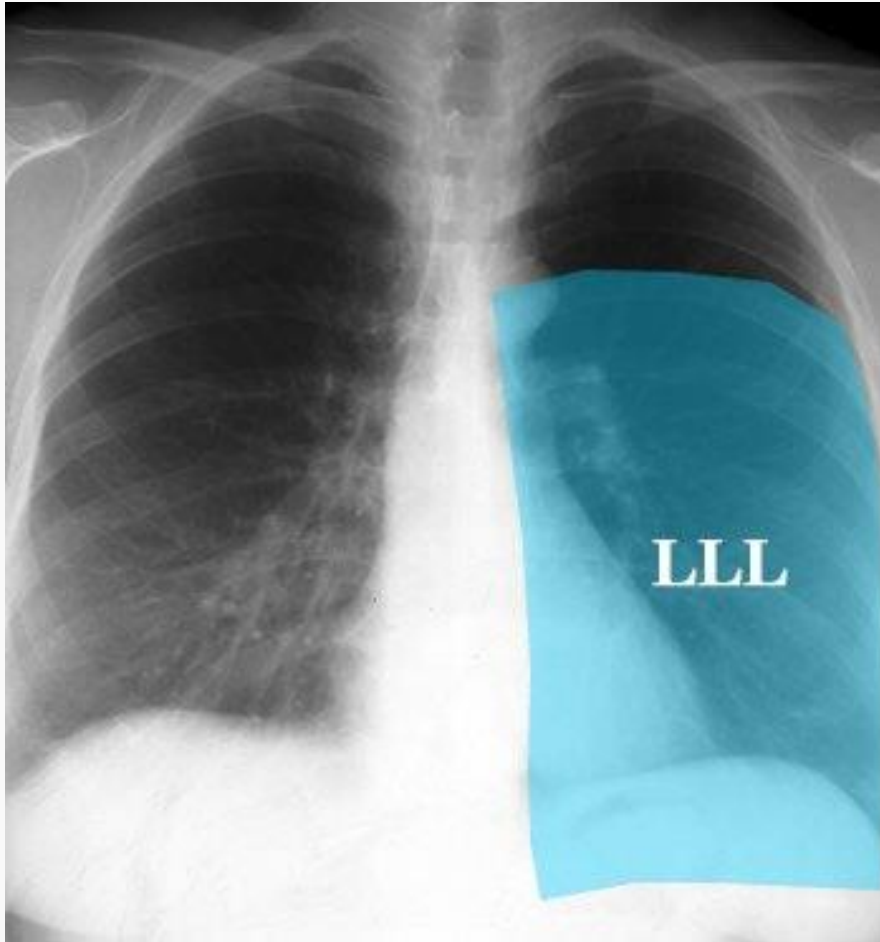


- LUL (Left Upper Lung)

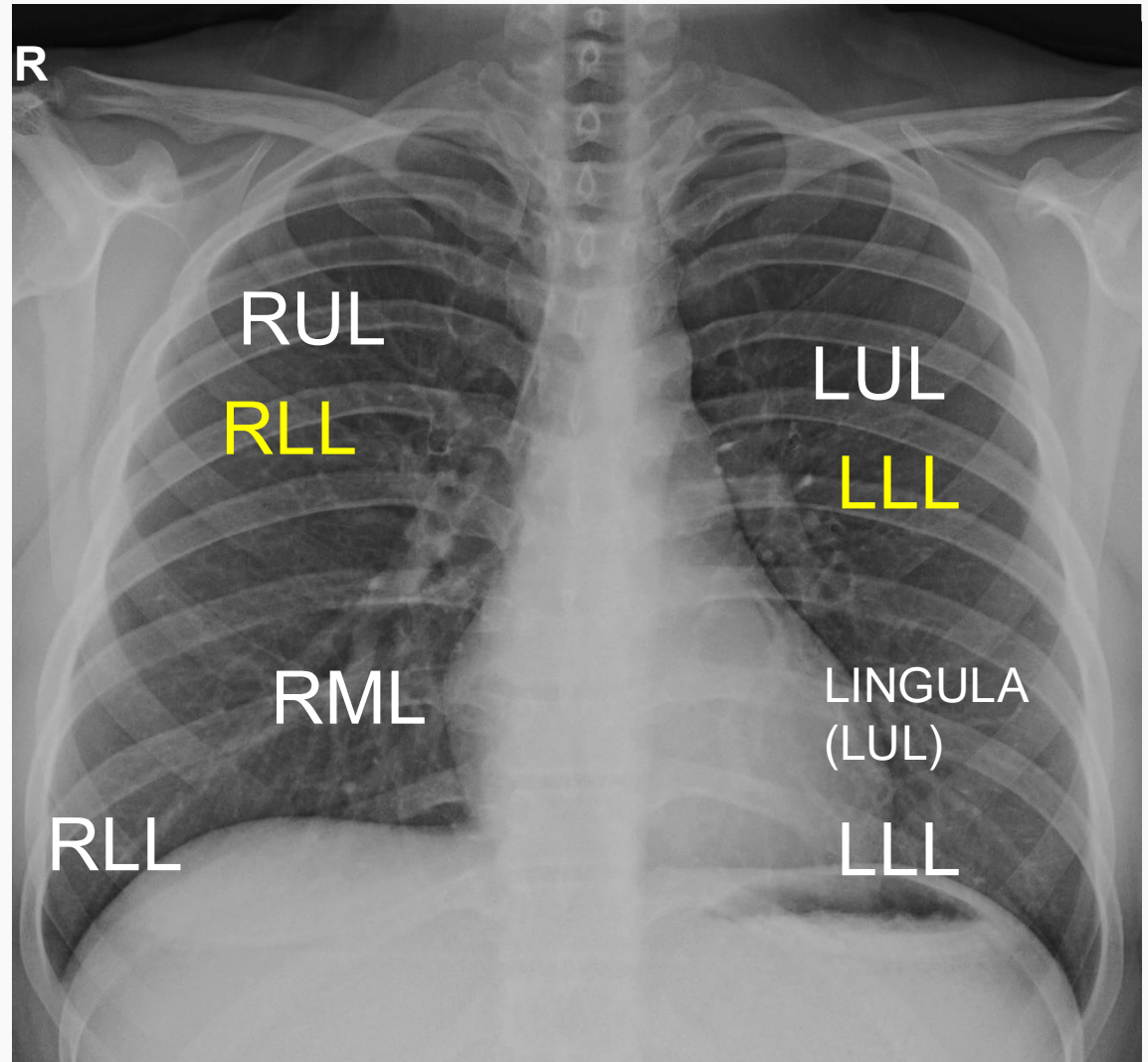


- Lingular lobe- adjacent to and allows visualization of the left heart border

- LLL (Left Lower Lung)

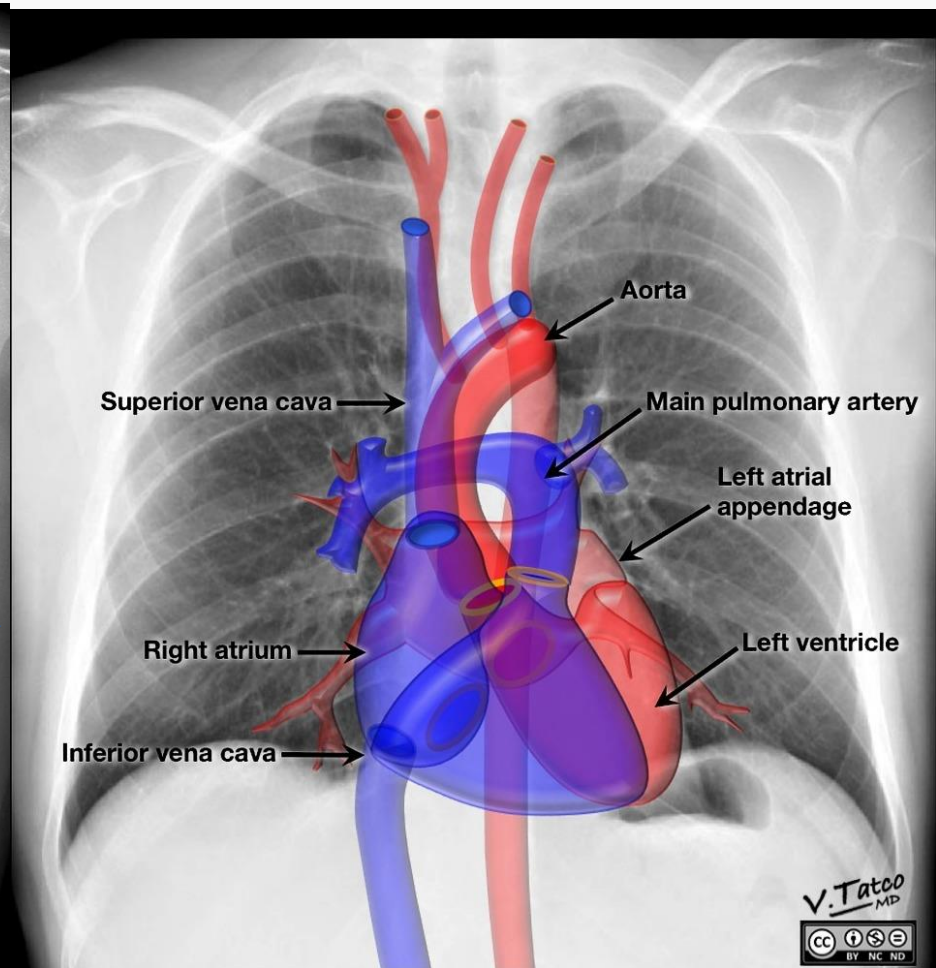
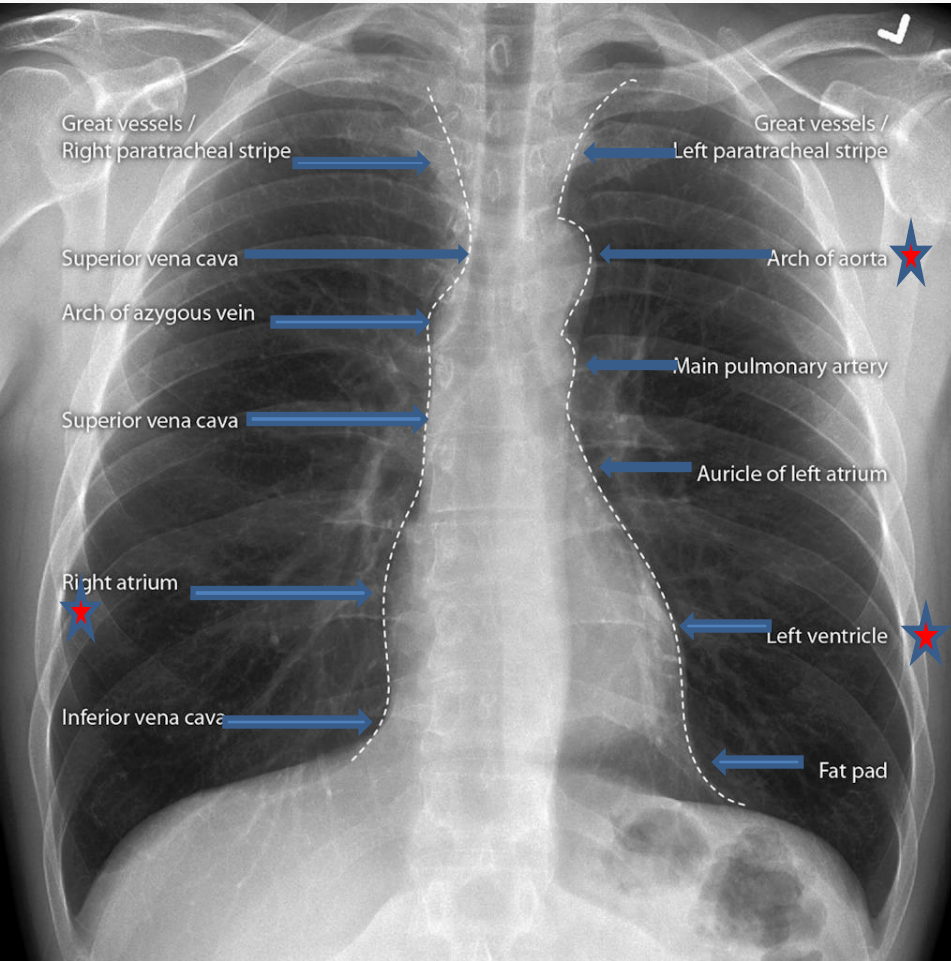


- RUL
 - RML
 - RLL
 - LUL
 - Lingula
 - LLL
-
- **SUPERIOR SEGMENT OF BOTH LOWER LOBES MAY PROJECT ABOUT AS HIGH AS AORTIC KNOB!!

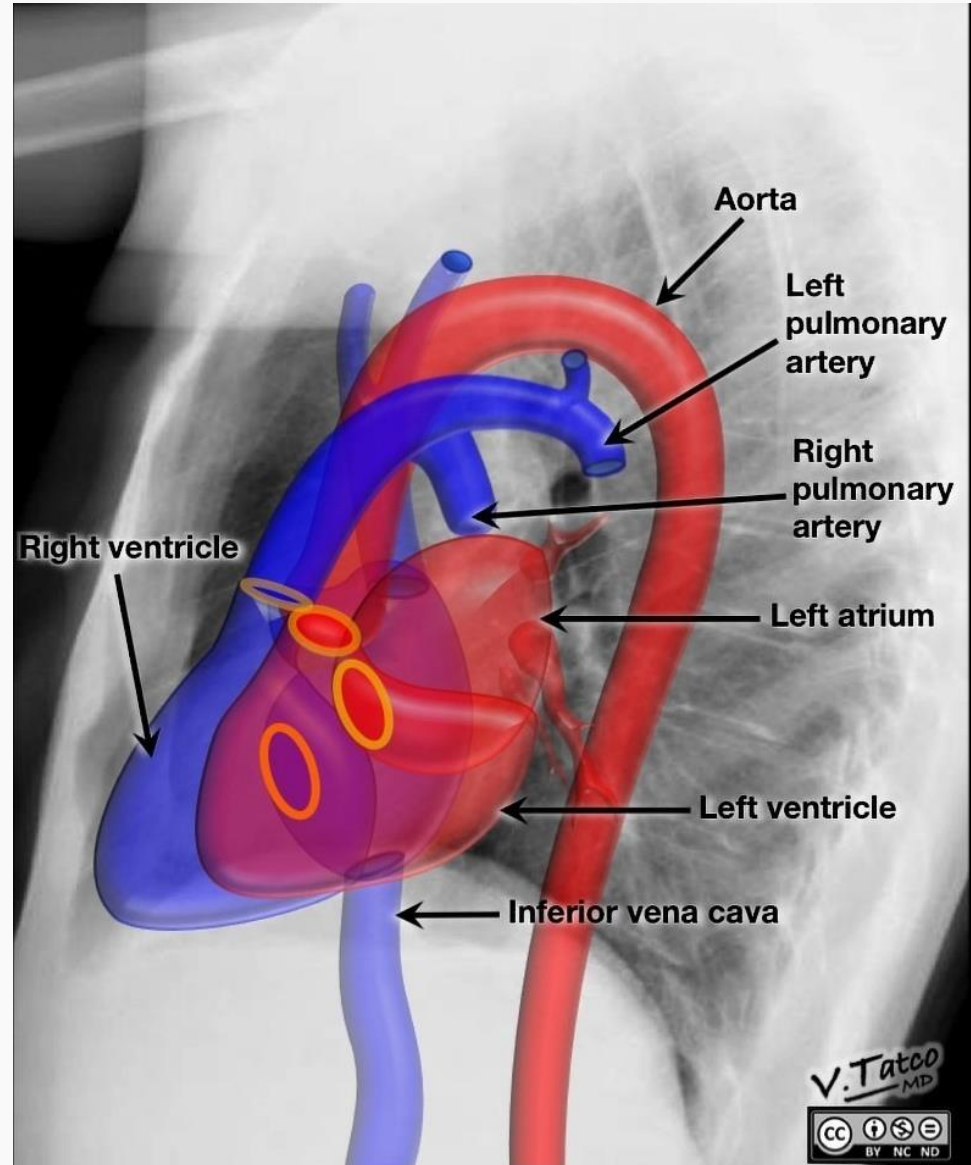
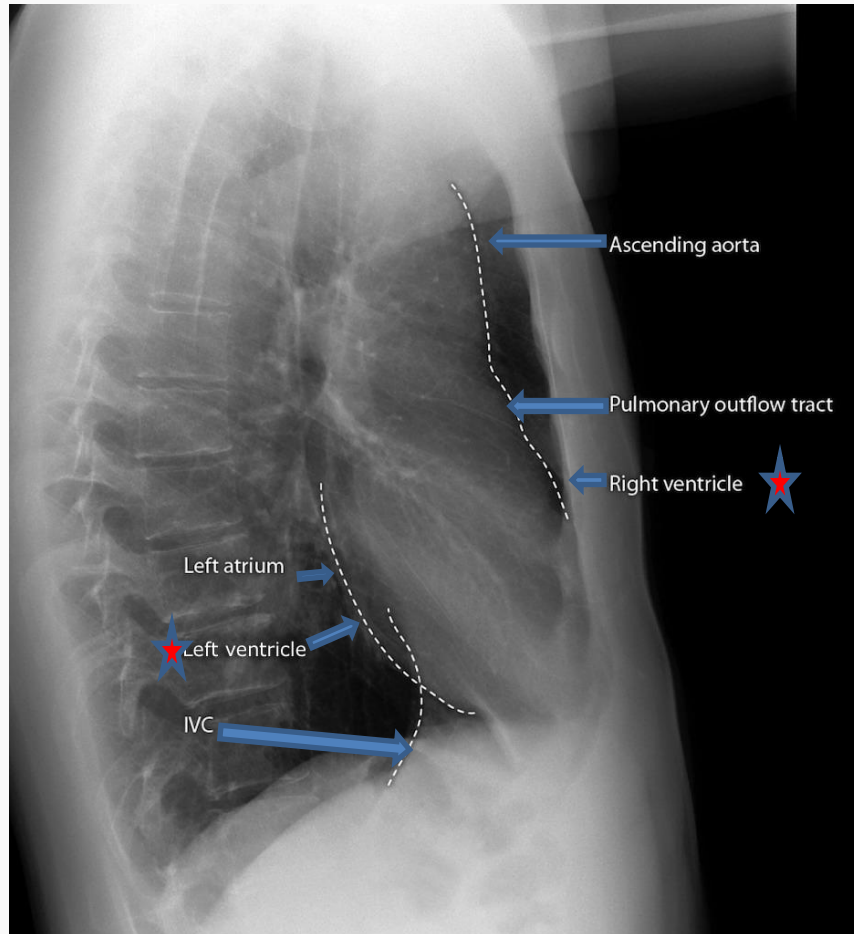


Normal Heart/Mediastinum

Normal Heart/Mediastinum



Normal Heart/Mediastinum



Case courtesy of Dr Vincent Tatco,
Radio
paedia.org. From the case <a
href="https://radiopaedia.org/cases/4
6331">rID: 46331

- The Key: A systematic approach

My approach:

- ▶ Assess study technique and quality
- ▶ Tubes and lines/Hardware/devices**
- ▶ Look at the partially imaged abdomen
- ▶ Bones and soft tissues
- ▶ Heart/Mediastinum
- ▶ Lungs and pleura- each one individually and then together for symmetry
 - ▶ OR
- ▶ **ABCDE method**



[http://www.meddean.luc.edu/lumen/
MedEd/Radio/curriculum/Pulmonary/St
ructure_Thorax_f.htm](http://www.meddean.luc.edu/lumen/MedEd/Radio/curriculum/Pulmonary/Structure_Thorax_f.htm)

Lecture Feedback Form:

<https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R>